

Human Capital, Competition and Mobility in the Managerial Labor Market*

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Abstract

We pose a structural model of the managerial labor market with general and firm-specific human capital accumulation, internal and external mobility, managerial bargaining power, and imperfect labor market competition. The model reconciles high pay with low mobility: outside offers can improve incumbent contracts without producing job switches. Firm-specific skill is central to this mechanism: it strengthens incumbent matches, raising poaching costs and weakening the leverage of a given outside offer. Competition therefore operates through renegotiation with incumbents: bargaining power alone would give managers 31.6% of match surplus, while competition lifts realized capture to 54.6%, and to 79.3% for externally hired CEOs. Making firm-specific skill portable shifts competitive gains from renegotiation toward mobility, increasing job switching, external CEO hiring, and manager welfare.

Keywords: managerial human capital, CEOs, the market for CEOs, executive pay, executive mobility, on-the-job search, structural estimation, human capital accumulation, firm-specific human capital

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1. Introduction

Is high CEO pay a reflection of managerial power or of competition for scarce managerial talent? Prevailing competitive theories, such as [Gabaix and Landier \(2008\)](#) and [Murphy and Zabojnik \(2007\)](#), attribute rising CEO pay to a market bidding for general managerial skill, while an influential alternative attributes it to managers' influence over their own pay-setting process ([Bebchuk and Fried, 2003](#)). One fact appears to settle the matter in favor of the second view: external CEO hires are rare, fewer than one percent of managers move laterally across firms in a given year, and the CEO labor market has grown more rigid over time ([Cziraki and Jenter, 2022](#); [Barry et al., 2025](#)). A market in which almost no one moves would seem to exert little discipline on pay. We argue that this inference is incomplete because it depends on the composition of managerial talent, and in particular on how much is specific to the manager's current firm.

In this paper, we pose and estimate a dynamic model of the managerial labor market. Managers accumulate general and firm-specific human capital. Over their careers, they receive job offers, including both internal promotions to CEO and external opportunities, and bargain with firms over pay. The framework allows us to quantify the roles of bargaining power, competition, and human capital and, importantly, how human capital interacts with competition in determining managerial pay and mobility.

How human capital shapes competition depends on whether those skills are portable. In our model, general skill is portable and valued by current and prospective employers, leaving the competitive balance unchanged. Firm-specific skill, by definition, is not portable: it raises a manager's value to her current firm but not to rivals. As it accumulates, outside offers become less competitive relative to her current job, making them less likely to induce a move and more likely to be matched by the incumbent. Firm-specific skill thereby redirects competitive pressure from mobility toward renegotiation and retention without reducing the pay she receives for the skill itself.

We estimate the model by indirect inference on a panel of managerial careers spanning 1992–2023, combined with hand-collected data on managerial tenure and labor market experience. Competition matters substantially despite low observed mobility: bargaining power alone would give

managers 31.6% of the surplus generated by their employment relationships, while competition lifts realized capture to 54.6%, so 42% of what managers capture is attributable to competition rather than to their own bargaining power. Most of this competitive pressure is exerted through renegotiation with the incumbent: roughly three quarters of career pay gains from improved contracts occur within the incumbent firm rather than through job switching. Our findings therefore contextualize the apparent tension between high managerial pay and low mobility: limited job switching need not imply weak competition or dominant managerial power.

Empirically, firm-specific human capital moderates this competition only slightly, but substantially changes its form. In a counterfactual that makes accumulated skill portable while holding its productive value fixed, managers' gains shift away from renegotiation with incumbent firms and toward job switching, and the share of CEOs hired from outside rises by about 14%. These changes in mobility and contracting raise managers' welfare by the equivalent of a permanent 0.59% increase in flow compensation on average, with substantially larger effects for managers who have accumulated the most firm-specific skill. Thus, high pay and low mobility can arise in a competitive market in which firm-specific skill redirects competition away from mobility and toward renegotiation with the incumbent.

While most of the executive labor market literature emphasizes firms' demand for talent, we take the manager's perspective and study how job opportunities, human capital, and bargaining power shape compensation and mobility over careers. We extend the equilibrium job search model of [Bagger et al. \(2014\)](#). Managers are employed either as executives or CEOs,¹ accumulate human capital while employed, and occasionally receive competing job offers. Our main contributions relative to [Bagger et al. \(2014\)](#) and the previous literature are to introduce firm-specific alongside general human capital and to allow internal and external CEO opportunities to arise through distinct frictions. These features help separate firm-specific skill from a potential preference for insiders ([He and Schroth, 2024](#)) in observed managerial mobility.

We model managerial compensation as a piece-rate contract: managers receive an endoge-

¹Throughout the paper, we refer to non-CEO executives as "executives."

nous portion $R \in [0, 1]$ of their contribution to firm output, following an important strand of the labor search literature (Postel-Vinay and Robin, 2002; Cahuc et al., 2006). This abstracts from specific contract forms, such as equity awards and cash bonuses, but delivers closed-form equilibrium contracts and allows bargaining power, labor market competition, and human capital to separately influence surplus capture. An attractive outside offer triggers bargaining between the manager, incumbent, and poacher, determining whether the manager switches firms or remains at a renegotiated contract. Labor market competition can thus generate stochastic, discrete pay increases even when the manager remains with her incumbent.

A key novelty of our model is that it characterizes how firm-specific human capital shapes labor market outcomes. Beyond inducing job lock, firm-specific skills also strengthen the incumbent firm’s incentive to match attractive outside offers, due to the rising match-specific productivity of managers over tenure. We show theoretically how firm-specific human capital affects whether labor market competition operates through renegotiated compensation or mobility across firms, speaking to the broader literature on human capital accumulation and matching in labor markets (Jovanovic, 1979; Felli and Harris, 1996; Moscarini, 2005; Bagger et al., 2014).

Our estimation produces several results. First, we find that both human capital accumulation and contractual improvements are important sources of managerial pay growth over careers. General human capital accumulates steadily with labor market experience, while firm-specific human capital accumulates more rapidly early in a manager–firm match, increasing the value of the incumbent relationship as tenure rises. Contractual gains are also quantitatively important: roughly three quarters of cumulative piece-rate growth is realized through within-firm contract revisions rather than through job switching. Thus, a substantial share of managerial pay growth occurs without mobility. These patterns also illustrate why reduced-form returns to experience and tenure need not measure human capital alone, as observed compensation reflects accumulated skill together with endogenous changes in contracts and labor market competition.

A natural question is whether these contractual gains reflect managers’ own bargaining power or competition for their services. Our estimation allows us to separate managers’ inherent bar-

gaining power from the additional surplus capture generated by competing job offers (following [Cahuc et al., 2006](#)). Competition is especially important for externally hired CEOs: poached CEOs capture roughly 79% of surplus, with 60% of that attributable to competition, compared with 50% and 36% for internally promoted CEOs. Thus, high realized managerial surplus capture need not imply correspondingly high inherent bargaining power.

We also analyze how firm-specific human capital shapes the form of labor market competition. In the portability counterfactual, we find that managers' contractual gains shift away from within-firm contract revisions and toward job switching: the share realized through switching rises by 4.9 percentage points (a quantitatively large 18% increase). Firm-specific human capital therefore changes the margin on which competition operates, redirecting competitive pressure away from mobility and toward the incumbent relationship.

This shift in the mobility margin also affects the composition of CEO appointments. When all human capital is portable, the consequent rise in external hiring is substantially larger among long-tenured managers, for whom accumulated firm-specific human capital is greatest. Firm-specific human capital therefore helps explain the predominance of internal CEO appointments, but it is not the primary source: internal CEO opportunities arrive more than four times as frequently as external CEO opportunities in the estimated model, so differences in opportunity-arrival frictions account for much of the remaining gap.

Finally, we quantify the welfare consequences of the non-portability of firm-specific human capital for managers. Our welfare measure isolates the labor-market cost of non-portability: it holds fixed the productive value of accumulated skill and changes only whether that skill is retained after a move. On average, job lock lowers manager welfare by the equivalent of a permanent 0.59% reduction in flow compensation. This cost reflects lower surplus capture and slower promotion into CEO positions, partially offset by managers remaining matched with more productive firms. The average masks substantial heterogeneity over the career. Measured this way, job lock raises welfare by 1.65% for managers with less than five years of tenure, when accumulated firm-specific skill is still limited and incumbent renegotiation remains valuable, but lowers

welfare by 2.51% for managers with ten to twenty-five years of tenure, when lost mobility and bargaining leverage dominate. Thus, the welfare cost of non-portability emerges as productive attachment to the incumbent firm increasingly restricts managers' outside options.

While our framework captures important features of the managerial labor market, we abstract from the information and incentive frictions studied in prior work. In particular, firms and managers have perfect information about match quality and human capital, so we do not capture asymmetric employer learning in internal versus external hiring (e.g., [Waldman and Zabochnik, 2025](#)), nor do we micro-found incentive pay in the presence of agency frictions such as moral hazard (e.g., [Lyman, 2024](#)).² Our focus is on equilibrium pay and mobility in a competitive managerial labor market, and on how human capital accumulation interacts with labor-market frictions.

Moreover, search in the model should be interpreted broadly as the arrival of employment opportunities, rather than literally as managers searching for vacancies. Differences in the arrival rates of internal and external opportunities may reflect executive search, preferences for insiders, or other search frictions. Importantly, the model separates these forces from firm-specific human capital at the level of primitives: promotion and mobility patterns discipline the opportunity-arrival processes, while compensation and employment choices conditional on opportunities discipline the accumulation and value of human capital. Thus, in our model, a high rate of internal promotion need not be attributed entirely to firm-specific skill and may also reflect broader frictions in the hiring process.

Literature Review. This paper contributes to several strands of literature. First, we engage with the literature on managerial human capital and CEO compensation, notably the debate on the relative importance of general versus firm-specific managerial skills. [Murphy and Zabochnik \(2004, 2007\)](#) model the firm's choice of an internal vs. an external candidate, and show that, in a market with an elastic supply of managerial labor, greater importance of general skill can rationalize the observed increase in executive pay. [Custódio et al. \(2013\)](#) provide an empirical

²Introducing asymmetric information and endogenous beliefs about match quality into our dynamic, multi-firm bargaining environment would substantially enlarge the state space and require tracking posterior beliefs over managers and matches, making estimation far more computationally demanding ([Adenbaum et al., 2024](#)).

proxy for general managerial skill and show that it is associated with higher total CEO pay, supporting [Murphy and Zbojnik \(2007\)](#). We provide a structural quantification of the relative importance of general and firm-specific skills.

Second, our paper contributes to the literature on managerial labor market dynamics and mobility ([Cziraki and Jenter, 2022](#); [Barry et al., 2025](#)). Prior work emphasizes information frictions and preferences for insiders as important explanations for low external CEO hiring ([He and Schroth, 2024](#)). We complement this literature by modeling managers as active participants in the labor market: managers choose among competing internal and external opportunities, accumulate human capital, and bargain with firms over pay. This allows us to quantify how firm-specific human capital affects both managers' mobility and firms' incentives to retain them. Our findings also relate to [Kecht et al. \(2025\)](#), who document greater reliance on externally acquired managerial experience at private and smaller public firms, consistent with heterogeneity across firms in the importance of general versus firm-specific managerial skill.

Third, we contribute to the literature on managerial rent extraction ([Bebchuk and Fried, 2003](#); [Taylor, 2013](#); [Coles et al., 2014](#)). Though existing studies recognize separate roles for bargaining power and labor market competition, they are often empirically indistinguishable. We use our model to separate these components and quantify their contributions to managerial rent extraction. While high managerial rents are often interpreted as evidence of agency frictions, our results show that they can arise endogenously from labor market competition.

Our paper also relates to the labor economics literature on human capital accumulation, job search, and mobility. Canonical models emphasize general human capital and on-the-job search ([Flinn, 2006](#); [Bagger et al., 2014](#)), while more recent work incorporates firm-specific human capital and studies its implications for training and mobility ([Lazear, 2009](#); [Lentz and Roys, 2024](#); [Gao et al., 2021](#); [Ma et al., 2023](#)). [Ferreira and Nikolowa \(2023\)](#) and [Waldman and Zbojnik \(2025\)](#) study managerial labor markets in which asymmetric employer learning and firm-specific human capital jointly favor internal candidates. We build on this literature by integrating general and firm-specific human capital with dynamic bargaining, allowing us to characterize how firm-

specific skill changes firms' responses to outside offers and the form through which labor market competition operates. This mechanism also relates to the organizational-capital literature, which emphasizes that firm-specific intangible assets may be embodied in key employees (Eisfeldt and Papanikolaou, 2013; Lustig et al., 2011).

Finally, job switching, renegotiation, bargaining, and the accumulation of general and firm-specific skills jointly generate endogenous relations between compensation, tenure, and experience that reduced-form wage models (Mincer, 1974; Abowd et al., 1999) may not capture, since unobservables such as labor market competition can confound inference. Our structural approach allows us to separately identify firm-specific human capital, general human capital, bargaining power, and labor market competition, and to quantify how these forces jointly shape managerial compensation and mobility.

The rest of the paper is organized as follows. Section 2 describes the data. Section 3 presents the model, and Section 4 discusses estimation and identification. Section 5 reports parameter estimates and our main results concerning the drivers of managerial compensation. Section 6 examines managerial bargaining power and labor market competition, and Section 7 studies the role of firm-specific human capital in mobility and compensation. Section 8 concludes. Model proofs are in Appendix A and further estimation details in Appendix C.

2. Data

We estimate our model on a panel dataset covering executive careers at US public firms from 1992 to 2023. Our primary data come from Execucomp, supplemented with manually collected information on when managers first joined firms and entered the labor market, extending the data collection in Gayle et al. (2015). We classify each manager in our sample as being employed in an executive (non-CEO) position, a CEO position, or unattached. A manager is classified as unattached when she leaves a firm in our sample for at least one year and subsequently reappears at another public firm; managers who permanently disappear from the sample are classified as

retired.³

Our measure of compensation is Execucomp TDC1, which includes each year’s salary, bonus, stock and option grants, long-term incentive payouts, and other non-equity incentive compensation. We deflate compensation to 2020 dollars before taking logarithms or constructing wage growth. Empirically, we treat annual TDC1 as the observed counterpart of the model’s annual compensation flow W ; it need not equal a directly observed contemporaneous share of firm output. We identify each labor market event for managers in our sample. Executives can be internally promoted to CEO, externally promoted to CEO, or move horizontally to an executive position at another firm. CEOs can move horizontally to other CEO positions.

Table 1 displays summary statistics (Panel A) and labor market transition statistics (Panel B), conditional on employment state (executive, CEO, or unattached). CEOs receive about 3 million at the median, compared with 958 thousand for executives. Wage growth averages about 9.8% per year for CEOs and 13.7% for executives. External CEO hires are uncommon relative to internal promotions: there are 667 external CEO hires (293 CEO-to-CEO moves and 374 external executive-to-CEO promotions), compared with 4,170 internal promotions. There are more executive-to-executive transitions (1,535), which are particularly useful for identifying general and firm-specific human capital because they provide pay changes around cross-firm job switches.

3. Model

3.1. Environment

The labor market consists of firms and managers. Time is continuous and both parties discount the future at rate δ . For a given manager, we let t denote their total experience in the managerial labor market and $\tau \leq t$ denote experience at their current firm (i.e., tenure). At any point in time, managers may find themselves in one of three employment states: if currently employed,

³We cannot perfectly distinguish retirement from permanent disappearance from Execucomp. In particular, a manager who leaves employment in our sample and never subsequently reappears is classified as retired even if she remains active elsewhere. Because terminal disappearance may therefore reflect disclosure rather than true labor-market exit, retirement is treated separately in the estimation. Section 4 describes its calibration and the construction of the non-retirement transition moments.

managers may hold a CEO position (C) or a non-CEO position (E), where we refer to the latter type of manager as an “executive.” Otherwise, managers may be “unattached” (U), in which case they do not currently hold a position in the managerial labor market. Managers have logarithmic utility over flow compensation.⁴

3.1.1. Production and Human Capital

A manager’s total human capital (in logs) is given by:

$$h = a + g(t) + k(\tau) \quad (1)$$

The parameter $a \sim N(0, \sigma_a^2)$ is a (publicly-known) manager-specific ability parameter reflecting permanent differences in individual productivity. The functions $g(t)$ and $k(\tau)$ denote general and firm-specific human capital accumulated through work experience, and are each cubic in their arguments. Importantly, any time a manager leaves their firm, general human capital $g(t)$ is retained while their firm-specific capital $k(\tau)$ is not: $k(\tau)$ resets to $k(0) = 0$ upon switching firms.

At each t , each manager-firm match generates output Y , with natural logarithm y :

$$y = a + g(t) + k(\tau) + f + \Delta_C \mathbf{1}\{\text{CEO}\} \quad (2)$$

where $f \in [f_{\min}, \infty)$ is a firm effect summarizing firm-specific characteristics which augment managerial productivity. This may include firm size, organizational culture, proprietary technology, etc. We henceforth refer to f as firm productivity which, as it is firm-specific, is redrawn from distribution Ψ any time a manager switches firms. The parameter $\Delta_C \geq 0$ is a CEO productivity premium: a manager occupying the CEO position generates additional log output Δ_C , reflecting the greater scope and responsibility of the role. It is convenient to define $p = f + k(\tau)$

⁴As we show below, logarithmic utility yields a tractable bargaining problem. Because pay is deterministic conditional on the match and there are no savings or state-contingent transfers, the specification introduces no risk-sharing motive.

as the manager’s match productivity, representing the portion of total productivity y tied to the current match, which is therefore lost upon switching firms. As we will later show, the dynamics of match productivity p play a critical role in shaping managers’ incentive to either remain in their current position or entertain outside offers.

While unattached, a manager’s tenure τ is fixed at zero and firm-specific human capital does not accumulate. It is important to note that unattachment is not synonymous with unemployment. Indeed, as discussed by [Cziraki and Jenter \(2022\)](#), unattached managers are largely those who temporarily exit employment within large, publicly-held firms in favor of other roles in, for example, government, non-profits, or privately-held firms. We therefore allow experience t , and thus general skills $g(t)$, to accumulate while unattached.

3.1.2. Labor Market Transitions and On-the-Job Search

Over their careers, employed managers encounter outside employment opportunities that arrive stochastically. We do not model managers as actively soliciting these offers; rather, an opportunity arises and the manager’s only decision is how to respond to it.⁵ For executives, three possible offers may be received: internal promotion to CEO, external promotion to CEO, or a lateral move to an executive role with another firm. Internal promotion opportunities arrive at Poisson rate λ_0 , while external offers for CEO and executive roles arrive at rates λ_1 and λ_2 , respectively. Crucially, internal promotions are distinct from external transitions in that firm-specific human capital $k(\tau)$ is retained following a within-firm move, but reset to zero following a cross-firm move.

For CEOs, the only viable outside offers are for CEO positions at different firms; we rule out demotions *ex ante*.⁶ As in the case of executives, we assume that external CEO offers arrive at

⁵This passive-arrival structure is well suited to the executive labor market. Because openly searching for outside employment may conflict with a sitting executive’s fiduciary obligations, we abstract from search intensity and let outside opportunities arrive exogenously at Poisson rates, interpreted as a reduced form of firms’ recruiting activity. The framework therefore retains the tractability of on-the-job search while remaining agnostic about executives’ ability to seek outside positions. Institutional features affecting recruiting load onto the estimated arrival rates: non-compete enforceability ([Shi, 2023](#)) restricts external mobility and thus outside-offer arrival, while succession-planning practices ([Cvijanović et al., 2023](#); [Celentano and Mello, 2024](#)) shape internal-promotion arrival.

⁶This assumption is motivated by the data. Most apparent demotions, in which a CEO becomes a non-CEO executive at the same firm, are advisory roles, either explicitly as an advisor to the new CEO or implicitly as board chair. A small number of true demotions remain. Because our model is not intended to capture temporary CEO appointments

rate λ_1 . We will show that despite this assumption, the rates of executive-to-CEO and CEO-to-CEO transitions will differ in equilibrium. As in all previous cases, CEOs forgo their firm-specific capital $k(\tau)$ upon taking a position with another firm. Whenever an outside offer arrives, the productivity f' of the competing firm is drawn from a distribution that depends on the manager's ability, $\Psi(\cdot | a)$, under which more able managers tend to meet more productive firms.

In addition to receiving job offers, managers may receive employment shocks inducing separation or exit from the labor market. We assume that managers retire and exit permanently at rate μ . Alternatively, at rate η manager-firm matches are dissolved and the manager enters unattached. While unattached, managers may potentially re-enter the labor market by receiving an offer for an executive or CEO position, which arrive at rates γ_E and γ_C , respectively.

3.2. Bargaining and Wage Contracts

Compensation takes the form of a piece-rate contract: managers receive an endogenous share $R \in (0, 1]$ of their marginal product Y . Output-linked pay, through bonuses and equity grants, is the dominant form of executive compensation, so we adopt an output-sharing contract and take this form as given.⁷ Consequently, log compensation is given by:

$$w = r + a + g(t) + k(\tau) + f + \Delta_C \mathbf{1}\{\text{CEO}\}. \quad (3)$$

Piece rates $r = \log(R) \leq 0$ are an equilibrium object determined via strategic bargaining between firms and managers, with $r = 0$ capturing the extreme case in which the manager fully extracts all surplus generated by the match. Throughout their careers, managers can increase their captured surplus r by leveraging competing job offers.

It is worth distinguishing at the outset the two forces that govern r . The first is bargaining power β , a structural primitive that fixes the share of newly created surplus—the gain above her

followed by demotion, we exclude these managerial spells from the estimation sample.

⁷We do not model the incentive problem that gives rise to performance-based pay, which is beyond the scope of our analysis; the piece rate r is the reduced-form share of output the manager captures. This follows an influential strand of the labor search literature (Postel-Vinay and Robin, 2002; Bagger et al., 2014).

threat point—that a manager captures in any single negotiation: her inherent, *ex ante* bargaining power. The second is competition, the rate and strength with which rival offers arrive, which raises a manager’s threat point and, through the sequential auction described below, bids her contract toward fuller extraction. Both forces raise r , but they operate through different channels: β governs the split within a negotiation, while competition governs how often and how favorably a manager’s outside option is refreshed. The two need not move together; in the limit of perfect competition, rival bids drive r to full extraction ($r = 0$) regardless of β . In that limit, competing bids raise the threat point to the full match payoff and leave no incremental surplus for β to divide: β becomes irrelevant to the allocation, not equal to one, though $\beta = 1$ would deliver the same allocation through a different channel.

When an offer arrives, a bargaining game is initiated between the manager and firm(s) bidding for their services, the outcome of which determines who wins the manager and at what contract. The bargaining protocol is an extension of [Cahuc et al. \(2006\)](#), the exact form of which depends on the type of offer in question: internal promotion, external promotion, or lateral move. Throughout, we let $V_i(r, \tau, f)$ denote the discounted value of employment for a manager in a type $i \in \{E, C\}$ role given state (r, τ, f) ,⁸ where τ is tenure and f is the current firm’s productivity.

3.2.1. Internal Promotions

Consider a manager employed in an executive position with state (r, τ, f) . Suppose their firm approaches them with a potential promotion to CEO. The firm offers piece rate r' characterizing the manager’s contract were they to accept the promotion, where r' satisfies the following condition:

$$V_C(r', \tau, f) = V_E(r, \tau, f) + \beta(V_C(0, \tau, f) - V_E(0, \tau, f)). \quad (4)$$

⁸Because general human capital $g(t)$ accumulates uninterrupted over the entire career, retained on every move and accruing even while unattached, it enters each value function only as an additive, experience-specific intercept that is common across a manager’s options and therefore drops out of every decision rule; we accordingly omit it from the state.

This rule encodes the two parties' outside options. The manager's outside option is to remain an executive at their current contract, worth $V_E(r, \tau, f)$; the firm's is to continue employing them in their current role.⁹ Relative to these outside options, the promotion creates surplus $V_C(0, \tau, f) - V_E(0, \tau, f)$, the joint gain from the change in title, measured between the CEO and executive match values at full extraction ($r = 0$) so that it reflects the change in the size of the match rather than in its division. The manager captures a share β of this incremental surplus, where β measures their bargaining power; the firm retains the complementary share $1 - \beta$.

Because the firm's fallback is to retain the executive, a promotion is executed in equilibrium only when it is mutually beneficial, that is, only when it raises the joint value of the match:

$$V_C(0, \tau, f) \geq V_E(0, \tau, f). \quad (5)$$

Match productivity p is unchanged by the promotion; where (5) holds, the transition is a change in job title that, by the manager's bargaining power, raises their captured surplus while leaving the firm weakly better off.¹⁰

3.2.2. Cross-Firm Transitions

For internal promotions, only two parties (the manager and incumbent firm) take part in negotiations. The situation is more nuanced for cross-firm job offers as negotiations entail three parties: the manager, the incumbent firm, and the poaching firm. Here, the two firms competitively bid for the manager's contract, while the manager exercises bargaining power over both. We let f' denote the productivity of the poaching firm, which is public information. The following proposition characterizes the three possible outcomes of this bargaining and the thresholds that separate them.

⁹A natural alternative is to let the firm's fallback upon a breakdown be to fill the CEO position with a next-best internal candidate rather than to retain the incumbent as an executive. We show in Appendix A.2.3 that this leaves the promotion rule (4) unchanged up to a relabeling of the manager's bargaining power parameter.

¹⁰This differs from the earlier draft, which measured the promotion surplus against the manager's current contract, $V_C(0, \tau, f) - V_E(r, \tau, f)$, and treated all internal offers as accepted. Measuring it instead against the executive's full-extraction value ensures that every executed promotion is weakly profit-improving for the firm.

Proposition 1. *Suppose a type- i manager with state (r, τ, f) receives an outside offer for a type- j position from a firm of productivity f' . There exist thresholds $\underline{\theta}_{ij}(r, \tau, f) \leq \bar{\theta}_{ij}(\tau, f)$ such that:*

- (i) *if $f' > \bar{\theta}_{ij}$, the manager leaves for the poaching firm;*
- (ii) *if $\underline{\theta}_{ij} \leq f' \leq \bar{\theta}_{ij}$, the incumbent retains the manager at a renegotiated contract; and*
- (iii) *if $f' < \underline{\theta}_{ij}$, the offer is ignored and the manager's contract is unchanged.*

The acceptance threshold $\bar{\theta}_{ij}$ is the poaching productivity at which the manager is indifferent between moving and remaining at full extraction,

$$V_j(0, 0, \bar{\theta}_{ij}) = V_i(0, \tau, f), \quad (6)$$

and the renegotiation threshold $\underline{\theta}_{ij}$ is the productivity below which no retention concession is warranted,

$$V_i(r, \tau, f) = V_j(0, 0, \underline{\theta}_{ij}) + \beta(V_i(0, \tau, f) - V_j(0, 0, \underline{\theta}_{ij})). \quad (7)$$

Moreover, the acceptance threshold $\bar{\theta}_{ij}$ is increasing in the current firm's productivity f .

Proof. See Appendix A.2.2. ■

The new firm's productivity f' determines whether the manager accepts an outside offer, is retained under a renegotiated contract, or ignores the offer entirely. The acceptance thresholds $(\bar{\theta}_{ij})$ reflect the fact that in all cases of cross-firm transitions, the match productivity of the new position must be high enough to compensate for the firm-specific capital $k(\tau)$ lost during the job transition. Additionally, the likelihood of accepting an offer declines as $k(\tau)$ increases, as firm-specific human capital accumulation increases match productivity p , gradually driving up the opportunity cost of switching firms. This induces *job lock*: imperfect transferability of human capital reduces the realized rate of cross-firm transitions.

Given the acceptance criteria, we next outline the bargaining outcomes in each instance. Throughout, we consider a manager employed in a position of type $i \in \{E, C\}$ with state (r, τ, f)

who receives an offer for a position of type $j \in \{E, C\}$. Suppose first that poacher productivity satisfies $f' > \bar{\theta}_{ij}$. The manager accepts the offer in this case, and their new contract r' satisfies:

$$V_j(r', 0, f') = V_i(0, \tau, f) + \beta(V_j(0, 0, f') - V_i(0, \tau, f)). \quad (8)$$

Upon accepting the type j offer, the manager's new value $V_j(r', 0, f')$ is the total surplus of their old match $V_i(0, \tau, f)$ plus a fraction β of the additional surplus generated by the new match. The value of accepting the competing offer reflects the fact that the manager will begin their new job spell with $k = 0$, so the match productivity at their new firm starts at f' . Further, the manager's starting rate r' depends explicitly on the productivity of their prior employer (Fee and Hadlock, 2003).

Additionally, note a subtle but important difference between the external and internal sharing rules. For internal promotions, the manager's outside option is $V_E(r, \tau, f)$, the value of remaining in their executive role at contract r . For external transitions, the manager's outside option is $V_i(0, \tau, f)$, the value of remaining in their current role at piece rate $r = 0$ (i.e., extracting full surplus). We show in Appendix A.2.2 that competition between the two firms drives the losing firm (the incumbent in this case) to bid full surplus $V_i(0, \tau, f)$ which, while never accepted in equilibrium, is a viable threat point which the manager can use as leverage. Managers are thus in more favorable bargaining positions when fielding external offers (Cahuc et al., 2006).

Next, suppose the manager receives an outside offer from a firm with productivity satisfying $f' \in [\underline{\theta}_{ij}, \bar{\theta}_{ij}]$. The manager's equilibrium response is to reject the offer and remain with the incumbent at a renegotiated contract. This renegotiated piece rate r' satisfies:

$$V_i(r', \tau, f) = V_j(0, 0, f') + \beta(V_i(0, \tau, f) - V_j(0, 0, f')). \quad (9)$$

Similar to the previous case, competition between the two firms drives the losing firm's bid (the poacher in this case) up to full surplus $V_j(0, 0, f')$. Again, while this is not accepted by the manager in equilibrium, the threat of acceptance induces the incumbent to favorably renegotiate their

contract to incentivize retention. Thus, outside competition may still benefit the manager even if they ultimately elect not to accept the new offer. It is, however, possible that a competing offer is not threatening enough to trigger renegotiation by the incumbent. In particular, outside offers satisfying $f' < \underline{\theta}_{ij}$ are rationally ignored. This condition states that, given equilibrium bargaining strategies, receiving an offer from a firm with productivity $\underline{\theta}_{ij}$ would leave the manager no better off than their current contract. As such, managers only consider offers with $f' \geq \underline{\theta}_{ij}$.

The interval $[\underline{\theta}_{ij}, \bar{\theta}_{ij}]$ contains the offers the incumbent meets with a counteroffer, retaining the manager at a renegotiated contract. We refer to it as the *renegotiation window*. It is where competition operates invisibly, repricing contracts without generating turnover. The following proposition describes how the window, the concessions negotiated within it, and the starting contracts of poached managers vary with tenure.

Proposition 2 (Firm-specific capital and outside competition). *Consider a type- i manager in state (r, τ, f) facing type- j outside offers, with thresholds $\underline{\theta}_{ij}(r, \tau, f) \leq \bar{\theta}_{ij}(\tau, f)$ as in Proposition 1. Suppose $k(\cdot)$ is weakly increasing; hold the contract r fixed, as it is between negotiations; and, when $i = E$, suppose the promotion surplus $S(\tau, f)$ is weakly increasing in τ .¹¹*

- (i) Both thresholds $\underline{\theta}_{ij}$ and $\bar{\theta}_{ij}$ are weakly increasing in tenure τ .
- (ii) The value gain from an offer of any fixed quality f' that triggers renegotiation is weakly decreasing in tenure τ .
- (iii) The starting piece rate r' of a manager poached by a firm of any fixed quality f' is weakly increasing in the tenure τ accumulated at the firm they leave.

Proof. See Appendix A.7. ■

The proposition characterizes how firm-specific capital shapes competitive dynamics over a job spell. Part (i) states that both thresholds shift rightward with tenure. Fewer outside offers are strong enough to poach the manager; offers that would once have done so are instead matched

¹¹The promotion surplus condition is verified at our estimated parameters.

by the incumbent, which retains the manager at a renegotiated contract. At the same time, more offers fall below the renegotiation threshold and pass without any response. Part (ii) states that the offers the incumbent does match command less: retaining a manager whose match has grown more valuable requires a smaller concession. Part (iii) applies at the moment the manager does move: the incumbent's willingness to pay for the capital the manager abandons sets the bid a poacher must beat, so the more capital left behind, the better the starting contract, consistent with the hiring premia documented by [Fee and Hadlock \(2003\)](#).

The proposition speaks to why the influence of competition on managerial pay cannot be gauged from observed poaching alone. General capital is portable, so its accumulation raises the manager's outside offers in step with the current match and leaves the terms of competition unchanged. Firm-specific capital, by contrast, raises the value of the current match but not the value a rival firm assigns to the manager. As it accumulates, outside offers lose ground relative to the match they are bidding against: a shrinking share can poach the manager and a growing share draws no response at all (part (i)), the offers the incumbent does match command smaller concessions (part (ii)), and their full force is felt only in the event of an actual poach, when the abandoned capital is priced into the winning bid (part (iii)). Firm-specific capital therefore weakens the competitive pressure under which the manager's share of the match is set, without reducing the manager's pay for the capital itself. Whether this leaves the labor market a minor influence on executive pay, or an influence that simply arrives through matched offers and retention raises rather than through moves, depends on the arrival rate of offers, their quality distribution, and the surplus each offer can unlock. These are the objects the estimation recovers.

3.2.3. Unattached Managers

Lastly, recall that unattached managers can re-enter the labor market with an executive or CEO job offer. For tractability, we follow [Bagger et al. \(2014\)](#) and assume that the value of unattachment V_U is equivalent to employment in the least-productive executive position, $V_U = V_E(0, 0, f_{\min})$, so that unattached managers accept any executive offer and any CEO offer whose value clears

V_U . Upon receiving an offer of type j from a type- f' firm, managers exit unattachment and begin their new role with piece rate r' satisfying:

$$V_j(r', 0, f') = \beta V_j(0, 0, f') + (1 - \beta) V_E(0, 0, f_{\min}). \quad (10)$$

3.3. Equilibrium Pay Process

For a manager of type $i \in \{E, C\}$, the equilibrium wage process implied by the bargaining rules above is

$$w_i = a + g(t) + k(\tau) + f + \Delta_C \mathbf{1}\{\text{CEO}\} + r_i^*, \quad (11)$$

where r_i^* is the realized piece rate following the manager's most recent negotiation. In the spirit of [Mincer \(1974\)](#), equilibrium compensation admits an additive decomposition into a manager fixed effect a , experience and tenure trends $g(t)$ and $k(\tau)$, a firm fixed effect f , a CEO premium Δ_C , and the endogenous piece rate r_i^* . The first five terms are familiar wage determinants; the last is what our framework adds, capturing the economics of search and bargaining.

The piece rate r_i^* captures the history-dependent effects of bargaining, outside options, and cross-firm competition on pay. Its key feature is that firm-specific capital, though forfeited from *output* when a manager moves, is not forfeited from *pay*: the value of the match she leaves strengthens her outside option, so the poacher must compensate her through a higher piece rate ([Lentz and Roys, 2024](#)). Firm-specific skill is thus converted from wage *level* into surplus *capture*. Starting pay consequently depends on the route into a position, with poached managers commanding higher piece rates than internal promotions, and poached sitting CEOs commanding more than executives poached into the same seat.

Between negotiations the piece rate is fixed; it moves only when a meaningful offer arrives, ratcheting toward fuller surplus capture as outside offers improve over a job spell and resetting when the manager moves. Firm-specific capital shapes these dynamics in two offsetting ways. On the intensive margin, by raising the value of the incumbent match, it reduces the concession needed to retain the manager against an offer; firm-specific capital secures retention “for free.”

On the extensive margin, by raising the productivity a poacher must offer to win her, it widens the range of offers resolved through renegotiation rather than mobility. Relative to portable skill, managers therefore receive more frequent but smaller contractual revisions, leaving the effect on realized renegotiation rents theoretically ambiguous, a question we take to the data.

Because r_i^* varies systematically with tenure and experience through firm-specific capital accumulation and destruction, selection into more productive firms, and competing offers, observed pay-tenure and pay-experience profiles conflate human-capital accumulation with endogenous bargaining and competition. Our structural approach resolves this by modeling the evolution of r_i^* explicitly, allowing us to recover the contribution of managerial human capital net of bargaining, competition, and outside options. The next section takes the model to the data, mapping the wage equation (11) to a Mincer-AKM decomposition of executive pay, transition rates to observed promotion and cross-firm mobility hazards, and bargaining power to the gap between tenure profiles of wage levels and wage growth.

4. Estimation

In this section, we describe the estimation algorithm, identification strategy, and target moments used to recover the structural parameters, finally assessing the model’s fit. We estimate via indirect inference (McFadden, 1989): we estimate a (possibly misspecified) auxiliary model on both observed and simulated data, and find parameters that minimize the distance between the empirical and simulated moments, weighted by the inverse variance-covariance matrix of the empirical moments, clustered at the manager level. Details are in Appendix C.

4.1. Empirical Implementation

We parameterize firm productivity using observed firm size and a size-orthogonal component of heterogeneity. Letting s_j denote standardized log firm size, we specify

$$f_j = f_{\min} + b_{\text{size}} \times s_j + v_j, \quad (12)$$

where b_{size} governs the pass-through of firm size to productivity and v_j captures residual firm heterogeneity. Following [Bagger et al. \(2014\)](#), we assume

$$\Psi(v) = 1 - \exp\left(-\left(\frac{v}{s_1}\right)^{s_2}\right). \quad (13)$$

We also allow for sorting between managerial ability and firm size. We divide managers into three equal-mass ability groups and allow the distribution of firms they encounter to depend on ability. For a manager in ability group q , with mean ability $\bar{a}_q$, the probability of encountering firm size s_j is proportional to

$$\exp\left(\rho_{\text{sort}} \times \frac{\bar{a}_q}{\sigma_a} \times \frac{s_j}{\sigma_s}\right), \quad (14)$$

normalized across firm-size levels. Higher values of ρ_{sort} therefore make higher-ability managers more likely to receive opportunities from larger firms. Sorting operates through firm size alone: conditional on size, the orthogonal component v_j is independent of managerial ability. This allows the model to match the positive association between managerial ability and employer size without conflating sorting with the effect of firm size on productivity. We estimate f_{min} , b_{size} , ρ_{sort} , and the Weibull scale and shape parameters s_1 and s_2 .

Labor market shocks (separation, job offers, retirement) occur according to the arrival rates specified in the model. When an internal or external opportunity arrives, we draw the relevant firm productivity and determine mobility and compensation according to the equilibrium protocol in [Section 3.2](#). Managers may also enter unattachment or retire, and unattached managers can subsequently return to executive or CEO employment.

General and firm-specific human capital evolve flexibly with experience and tenure. We parameterize both using cubic functions,

$$g(t) = \sum_{q=1}^3 \delta_g^q t^q, \quad k(\tau) = \sum_{q=1}^3 \delta_k^q \tau^q, \quad (15)$$

and estimate the two sets of coefficients jointly with the remaining structural parameters. We

constrain both profiles to remain nonnegative over the computational grid.

Given managers' states, we compute model-implied compensation according to (11). For estimation, we allow observed log compensation to differ from this model-implied value through an additive, persistent residual,

$$w_{it}^o = w_{it} + \epsilon_{it}, \quad \epsilon_{it} = \nu_e \epsilon_{it-1} + e_{it}.$$

This residual captures sources of variation in measured compensation that are omitted from the model's parsimonious state space.¹² We assume that $e_{it} \sim N(0, \sigma_e^2)$ and jointly estimate ν_e and σ_e alongside the previously defined structural parameters.

We simulate managerial careers forward to approximate the model's stationary distribution, following, e.g., Bobba et al. (2021), and construct the same auxiliary statistics from the resulting simulated panel as in the observed data. Appendix C.1 reports the initialization, burn-in period, panel dimensions, state-space discretization, calibrated inputs, and complete simulation protocol.

4.2. Identification Strategy

All parameters are estimated jointly. We organize the identification discussion around the mobility, wage-level, and wage-growth moments that provide the primary identifying variation for different features of the model. Section 4.3.1 and Appendix C.7 provide statistical sensitivity evidence for these mappings.

4.2.1. Labor Market Mobility

Managers may transition among unattachment, executive employment, and CEO employment through separation, horizontal moves, internal and external promotions, and re-employment.

¹²The process ϵ_{it} is an econometric residual, not a source of compensation risk faced by managers: it enters neither utility nor the bargaining problem and therefore perturbs simulated compensation moments without affecting managers' choices or transition hazards. It may capture measurement error and omitted determinants of observed compensation, including noise in grant valuation and financial reporting. Because the residual follows the manager across employers, it cannot capture firm-specific capital, which accumulates with tenure and is lost upon switching. Adding such residual variation is common in empirical implementations of labor-search models, in part to avoid econometric singularity (see, for example, Eckstein and Wolpin, 1990; Flinn, 2006).

We calibrate the retirement rate μ to 0.0619, which matches average lifetime managerial experience of 16.16 years in the estimation sample. To identify the remaining transition rates, we fit Kaplan–Meier survivor functions by employment state and transition type. We match transition probabilities within the first five years of tenure and between five and ten years; for transitions from unattachment, we target re-employment within two years. Appendix C.2 details the construction of these moments. Because realized transition probabilities depend endogenously on employer productivity and tenure-varying match quality, these moments also discipline firm heterogeneity and human capital. We additionally target the share of managers employed as CEOs, which disciplines the net flows into and out of CEO employment.¹³

4.2.2. Compensation Levels, Growth, and Bargaining Power

Managerial compensation reflects experience, tenure, manager and firm heterogeneity, and a dynamically negotiated piece rate. Because the piece rate—the channel through which labor-market competition affects pay—is unobserved, we choose parameters to jointly match misspecified reduced-form regressions for compensation levels and growth in the observed and simulated data. These moments distinguish human capital, heterogeneity, and bargaining dynamics.

For manager i in fiscal year y , we estimate the following Mincer–AKM wage regression (Mincer, 1974; Abowd et al., 1999):

$$w_{iy} = g_{rf}(t_{iy}) + k_{rf}(\tau_{j(i,y)}) + \theta_C \text{CEO}_{iy} + \zeta_{j(i,y)} + \xi_i + u_{iy}, \quad (16)$$

where w_{iy} is log real total compensation, $j(i, y)$ is the employer, g_{rf} and k_{rf} are cubic reduced-form profiles in experience and tenure, and $\zeta_{j(i,y)}$ and ξ_i are firm and manager fixed effects. The remaining terms are the CEO indicator and residual.

¹³We construct these moments using managers with at least one observed non-retirement transition. Because Execucomp disclosure is limited to the CEO, CFO, and three other most highly compensated executives, disappearance from the sample need not represent retirement or labor-market exit. We therefore do not treat terminal sample attrition as zero promotion or separation hazards, and apply the same selection rule and transition-moment construction to the simulated careers. Consistent with this restriction, we target the CEO employment share in the estimation sample, approximately 34%, rather than the 20% share in the full Execucomp sample.

Because experience persists across employers while tenure resets, movers and stayers separately identify $g(t)$ and $k(\tau)$. The mean, variance, and skewness of the firm fixed effects identify $f_{\min}$ and the scale and shape of the firm-productivity distribution.¹⁴ The variance of the manager effects disciplines ability dispersion σ_a , the CEO coefficient identifies the CEO productivity premium Δ_C , and the variance and autocorrelation of u_{iy} discipline σ_e and v_e . We additionally target slopes from two regressions on employer size, both estimated across manager-year observations: the firm-effect slope pins down b_{size} , while the manager-effect slope pins down ρ_{sort} .

Structurally, compensation growth is the first difference of the model's compensation equation and therefore contains the same primitives as compensation levels. Matching its reduced-form is nevertheless informative because the omitted piece rate loads differently on levels and changes. Focusing on managers with at least two years of consecutive employment, we estimate

$$\Delta w_{iy} = \Delta g_{rf}(t_{iy}) + \Delta k_{rf}(\tau_{iy}) + \theta_{\Delta C} \Delta \text{CEO}_{iy} + \zeta_{j(i,y)} - \zeta_{j(i,y-1)} + u_{iy}^{\Delta}, \quad (17)$$

where Δg_{rf} and Δk_{rf} are the year-over-year changes in the experience and tenure polynomials, and ΔCEO_{iy} is the change in CEO status. The current and previous firm effects absorb employer wage premia at job switches.

Experience and tenure advance together within job spells; at switches, experience continues while tenure resets. Jointly matching the level and growth profiles therefore distinguishes human capital processes with similar compensation levels but different within- and across-job dynamics. We also target the variance and autocorrelation of u_{iy}^{Δ} , which further discipline σ_e and v_e .

Bargaining power affects wage levels and growth differently. The level regression records the current contract, whose piece rate embodies the cumulative effect of past revisions. The growth regression instead records the current change in the piece rate, so bargaining power affects wage growth through revisions that occur in that year. Firm premia have a distinct loading: the level regression absorbs the current employer's premium, while the growth regression absorbs the

¹⁴Limited mobility introduces estimation noise into the empirical firm effects. We therefore target noise-corrected variance and skewness; Appendix C.3 details the split-sample correction.

change in employer premia at a job switch. Firm-specific human capital, by contrast, enters both profiles directly. Jointly matching the two profiles therefore separates bargaining power from both firm-specific human capital and firm premia. Appendix C.7.2 formalizes this argument and shows that the level–growth variation responds strongly to bargaining power but minimally to other parameters. Importantly, we do not interpret the tenure profile of compensation alone as evidence of renegotiation, as other contracting mechanisms can also generate persistent pay growth (e.g., [Shue and Townsend, 2017](#)); identification instead comes from jointly matching compensation levels and changes with mobility and promotion patterns.

Moreover, pay changes around transitions provide further variation that helps separate bargaining power from outside-option-driven wage changes. Internal promotions make this distinction especially transparent: the manager changes roles and renegotiates within the same firm, so there is neither a change in the firm premium nor a contemporaneous rival bid. The estimation itself treats promotion types symmetrically: the same change-in-CEO-status coefficient captures pay changes at internal and external CEO promotions, and all job switches contribute to the wage-growth moments. At external transitions, the mobility moments discipline the outside-offer process, while the firm effects absorb changes in employer premia.

4.3. Model Fit

Labor market mobility. Panel A of Table 2 reports job transition probabilities and the share of managers employed as CEOs. Internal promotions are by far the most common transitions into CEO employment, a ranking the model reproduces, and external CEO hiring is matched closely. The main misses are early unattachment (−1.1 percentage points), late internal CEO hiring (−1.8 percentage points), early external executive hiring (−0.9 percentage points), and the CEO share (+2.9 percentage points). These gaps are moderate economically and concentrate where fixed arrival rates must jointly fit early and late transitions.

Auxiliary regressions and human capital profiles. Panels B and C of Table 2 report moments from the Mincer wage regression in (16) and the wage-growth regression in (17). Fit is

strongest for the firm and manager effects and size slopes, with modest misses in the CEO coefficients and wage-growth residual dynamics. The two CEO coefficients must be generated by the same CEO premium, bargaining mechanism, and residual process. The higher-order human-capital coefficients are precisely estimated and strongly correlated, and the optimal weighting matrix accounts for this covariance by evaluating them jointly. Individual coefficient discrepancies therefore need not imply poor profile fit: Panels A and B of Figure 1 show that the reduced-form profiles align closely. Panel C shows why these profiles should not be interpreted as structural accumulation: job moves reset tenure while raising pay through competition, biasing reduced-form experience upward and tenure downward.

Wage dynamics around CEO appointments. Figure 2 provides a demanding untargeted validation exercise. It plots event-time coefficients from manager×firm fixed-effects regressions of log compensation around both internal promotions to CEO and external moves into CEO positions, estimated identically in the observed and simulated panels. Although these event-time moments do not enter the objective, the model reproduces the large compensation increase at appointment and its persistence over the subsequent five years. The model somewhat understates the impact-year increase and does not reproduce the pronounced pre-appointment wage run-up. The pre-trend is consistent with anticipatory compensation or endogenous transition timing, mechanisms absent from the model’s exogenous opportunity processes.

4.3.1. Sensitivity of Parameters to Moments

Following Andrews et al. (2017), we compute the sensitivity of each parameter to each target moment, evaluated at the optimal parameter vector. Transition parameters respond most strongly to the corresponding hazards, the tenure reset at job switches separates firm-specific from general human capital, and the level–growth wedge disciplines bargaining power. Appendix C.7 reports the full sensitivity matrix.

5. Estimated Career and Compensation Dynamics

The next three sections present our main results. This section characterizes the estimated opportunity, human capital, and contractual processes shaping managerial careers and compensation. Section 6 separates bargaining power from labor market competition, and Section 7 isolates the consequences of nonportable firm-specific human capital.

5.1. Internal and External Labor Market Opportunities

Table 3 reports the parameter estimates. Our estimates of λ_0 (0.0295) and λ_1 (0.0070) imply that internal CEO opportunities arrive at roughly four times the rate of external CEO opportunities. This disparity mirrors the predominance of internal promotion in the data (Cziraki and Jenter, 2022) and may reflect several reinforcing frictions, among which we do not attempt to distinguish.¹⁵ Lateral offers of executive positions at other firms arrive at a higher rate ($\lambda_2 = 0.0368$), consistent with an active cross-firm market for executive talent (Cai et al., 2015). Together with incumbent retention, this opportunity structure generates long tenures and few cross-firm moves, shaping the human capital managers accumulate, the firms they match with, and the form that competition for their services takes.

5.2. Human Capital and the Allocation of Managers to Firms

Panel C of Figure 1 plots the estimated human capital processes. General human capital accumulates nearly linearly over a manager’s career. Firm-specific human capital is more front-loaded, accumulating early in a match and peaking after roughly sixteen years of tenure. To illustrate, consider two equally able managers with fifteen years of experience, one with fifteen years of tenure and one with five. The long-tenured manager’s human capital contribution to productivity is about 37% larger, but more of it is firm-specific. If both moved, she would retain only 47% of

¹⁵These include non-compete agreements (Shi, 2023), external search costs (Geelen and Hajda, 2024), tournament incentives (Agrawal et al., 2006), information frictions (Capron et al., 2024), and non-pecuniary preferences for insiders (He and Schroth, 2024).

her current human capital productivity, compared with 64% for the recent mover.¹⁶ Firm-specific capital therefore raises both the value of retention and the cost of poaching.

Heterogeneity in firm productivity is large: a position at the 90th percentile of the estimated distribution is roughly 5.9 times as productive as one at the 10th. Who holds these positions reflects two forms of sorting. First, the offer tilt ρ_{sort} directs large-firm opportunities toward high-ability managers, but realized sorting remains incomplete because offers arrive at finite rates, moves destroy firm-specific capital, and compensation is bargained. In frictionless assignment models, by contrast, perfect sorting arises by construction and the distribution of managerial skill is inferred from observed pay and firm size given perfect matching (Gabaix and Landier, 2008; Terviö, 2008). Here, ability remains a meaningful determinant of who runs the largest firms, even though the market falls well short of that benchmark.

Second, managers sort by experience. Firm productivity does not depend on experience, and general human capital, being portable and common across alternatives, drops out of mobility thresholds. Experienced managers nonetheless reach more productive firms because they have had more opportunities to draw and accept better matches, so the most productive firms are on average run by the most seasoned managers with the largest stocks of general skill (Figure B.1). This joint progression offers a market-based reading of the generalist pay premium (Custódio et al., 2013): experienced managers earn more both because general capital is priced and because they have had more opportunities to move up the firm distribution.

5.3. Decomposing Managerial Compensation

Panel A of Table 4 decomposes the variance of log compensation into permanent ability, general and firm-specific human capital, firm productivity, the piece rate, and the measurement residual. Permanent ability accounts for 47% of pay dispersion. This share is in line with reduced-form evidence on the importance of manager fixed effects in executive pay: Graham et al. (2012) attribute 54% and 5% of explained pay variation to manager and firm fixed effects, respectively. Because

¹⁶These calculations convert the estimated log human capital profiles into productivity levels. Table B.1 reports the corresponding variance decomposition of accumulated human capital into general and firm-specific components.

sorting is imperfect, the model carries manager and firm heterogeneity separately rather than inferring managerial skill jointly from pay and firm size under perfect matching (Gabaix and Landier, 2008; Terviö, 2008). Firm productivity accounts for 15%, and the covariance between ability and firm productivity, the equilibrium sorting term, contributes another 10%.¹⁷

Permanent ability explains persistent differences across managers, but not changes in compensation over careers. Panel B of Table 4 conditions on ability and reports the remaining components, which together account for 48% of total pay variance. The piece rate explains 38% of this net-of-ability dispersion, compared with 31% for firm productivity, and about 10% for human capital overall. The piece rate also covaries negatively with firm productivity and ability: managers with larger underlying productivity components tend to receive smaller shares of output.

Piece-rate dispersion is largest early in managers' careers, accounting for 70% of net-of-ability variance in the first five years and 44% among managers with ten to twenty-five years of experience. Negotiated contracts are therefore a major, though not exclusive, source of career pay dispersion. Unlike human capital, which accumulates continuously with experience and tenure, the piece rate changes only when a contract is repriced, so its dispersion at any point in a career reflects the sequence of repricings that preceded it.

5.4. Contractual Returns: Competition without Mobility

The cumulative change in the piece rate, which we call contractual returns, arises through two channels: moving to a firm offering a higher piece rate or revising the incumbent contract. Panel A of Figure 3 decomposes these returns over experience: contract revisions account for roughly three quarters, and job switching for the remainder. Panel B shows contractual returns over uninterrupted tenure, where all gains arise through incumbent revisions by construction. By twenty years of tenure, piece-rate growth reaches roughly 0.75 log points without a job switch, about equal to the contribution of firm-specific human capital over the same period. Most contractual gains therefore occur without a firm change. Outside-offer retention is one mechanism generat-

¹⁷The measurement residual accounts for 6.9% of total pay variance and 14.5% of the variance remaining once permanent ability is removed.

ing this pattern, so low mobility need not imply weak competition.

We do not interpret the model’s discrete, offer-triggered renegotiation mechanism literally. Annual pay reviews, retention grants, and peer benchmarking may transmit information about a manager’s outside market (Bizjak et al., 2008; Faulkender and Yang, 2010); model-implied renegotiations capture the component of these within-firm revisions that responds to that market. The key restriction is that competition can reprice a contract without a move, allowing intense competition to coexist with the near-absence of cross-firm mobility.

The size of each contractual adjustment depends on the productivity of the losing bidder. A more productive incumbent raises the concession a poaching firm must make, consistent with the finding of Fee and Hadlock (2003) that hiring grants for outside managers increase with the productivity of the previous employer, while a more productive poacher raises the concession the incumbent must make to retain her. Continued offer arrival therefore generates contractual pay growth even at long tenures, when accumulated firm-specific capital makes switching costly.

Each such adjustment reflects both labor market competition and managerial bargaining power. Because within-firm revisions accrue to managers who remain with the firm, they can resemble rents generated by weak governance, and reduced-form data cannot separate the two. Section 6 uses the model to do so.

6. Managerial Bargaining Power and Labor Market Competition

Our estimate of β (0.3164) implies that managers’ ex ante bargaining power is on par with estimates for ordinary workers (Bagger et al., 2014). Their *realized* surplus capture is nonetheless much higher, with the wedge reflecting labor market competition in our model. Separating the two matters for corporate executives in particular, as high managerial rents are often read as evidence that managers exercise undue influence over their own pay (Bebchuk and Fried, 2006).¹⁸ Because our model makes outside competition explicit, we can separate the two, following Cahuc

¹⁸Prior work finds managers capture substantial shares of surplus. Taylor (2013) estimates that CEOs capture about half of the surplus generated by good news about their ability, and notes that this share reflects both parties’ outside options in the labor market; Page (2018) and Barry (2024) estimate that boards place large weights on CEO utility, pushing pay above value-maximizing levels.

et al. (2006); to our knowledge this wedge has not previously been quantified for corporate executives. We find that competition's contribution is substantial and varies with tenure, experience, and whether a CEO was promoted from within or hired from outside.

6.1. Decomposition of Managerial Surplus

Following Cahuc et al. (2006), we compare the share of match surplus managers actually capture with the share they would capture from bargaining alone. A manager who never fielded a competing offer, the case $\lambda_1 = 0$ in our model, would bargain only against unattachment, and would be paid that outside option plus the fraction β of the surplus her match creates.¹⁹ More generally, for a CEO in a type f firm with tenure τ and experience t , pay can be written as

$$w = f_{\min} + b(r, p) (p + \Delta_C - f_{\min}) + a + g(t) \quad (18)$$

Here, $p + \Delta_C - f_{\min}$ is the surplus a CEO's match creates over unattachment, and the share $b(r, p)$ is the fraction of it that the manager's pay reflects.²⁰ Equating (18) to the wage our model implies, ability a and general capital $g(t)$ cancel, leaving:

$$b(r, p) = 1 + \frac{r}{p + \Delta_C - f_{\min}}. \quad (19)$$

Note that $b(r, p) \in [\beta, 1]$, so the bias from neglecting competition is weakly positive.²¹ Thus, within this decomposition, we use labor market competition to refer specifically to the effect of outside offers, holding the bargaining parameter β fixed. We then decompose average realized

¹⁹Recall that in our model, unattachment has the same value as employment in the least productive firm, $f_{\min}$.

²⁰We state the derivation for CEOs; it is analogous for executives, whose match productivity omits the CEO premium and whose outside opportunities include lateral moves and internal promotion.

²¹If $r = 0$, then $b = 1$: the manager captures the full match surplus. At the other extreme, in the no-competition benchmark of this exercise ($\lambda_1 = 0$), bilateral bargaining against the unattachment threat point delivers $r(p, f_{\min}) = -(1 - \beta)(p + \Delta_C - f_{\min})$ and hence $b = \beta$ exactly. Cross-firm competition raises r within this range, so realized capture rises from β toward 1 while the primitive β is unchanged.

surplus capture, $\bar{b} \equiv \mathbb{E}[b(r, p)]$, using the simple identity:

$$1 = \underbrace{\frac{\beta}{\bar{b}}}_{\text{baseline bargaining component}} + \underbrace{\frac{\bar{b} - \beta}{\bar{b}}}_{\text{competition component (above baseline)}} \quad (20)$$

We estimate $b(r, p)$ in our simulated sample (Table 5). In the full sample, realized surplus capture is 54.6%, close to the roughly 50% share of good-news surplus that Taylor (2013) estimates CEOs capture, an object that, like $\bar{b}$, reflects both competition and bargaining power. The decomposition in (20) attributes roughly 42% of observed surplus capture to labor market competition and 58% to baseline bargaining power. Thus, despite low CEO cross-firm mobility, competition substantially affects pay. The two facts are compatible because pay is repriced only when an offer arrives and, once improved, never falls back: between offers the contract is frozen while the match appreciates, leaving infrequent offers substantial surplus to unlock. Competition therefore operates through consequential repricings rather than turnover.

Subsample estimates of $b(r, p)$ show that this bias varies systematically across observables. Panel A of Table 5 shows that it is largest among poached CEOs: competition explains roughly 60.1% of surplus for external hires, versus 36.2% for internal hires. Poaching initiates a bidding war with the incumbent that the manager can leverage into a more favorable contract; internal bargaining has no such direct competitive pressure.

Panel B shows that competition's share of surplus rises with experience. Because piece-rate contracts are history-dependent, gains from a bidding war persist, so competitive gains accumulate over a career. The tenure profile runs the other way: competition's share falls from 51.2% in the first five years at a firm to 38.3% beyond ten. Two forces drive this decline. First, low-tenure cells contain more recently poached managers; externally hired CEOs capture 79.3% of surplus, compared with 49.6% for internal promotees (Panel A). Second, as firm-specific capital accumulates, outside offers lose ground against the incumbent position: fewer trigger a response, and those the incumbent matches generate smaller raises (Proposition 2). Taylor (2013) conjectures this force, noting that CEOs' surplus share may fall with tenure as their human capital becomes

firm specific; our estimates give the conjecture quantitative content. Thus, competitive gains accumulate over a career even as the influence of outside offers weakens with tenure. The next section shows how firm-specific human capital, by changing the value of incumbent and outside matches, generates this pattern.

7. Firm-Specific Human Capital, Mobility, and Managerial Rents

The estimated model reconciles two seemingly conflicting facts. Managerial mobility is low: most CEOs are promoted internally and spend much of their careers at a single firm. Yet Table 5 shows that CEOs capture a large share of surplus, with cross-firm competition playing an important role. The two are consistent because competition need not produce moves to matter: roughly three quarters of managers' contractual returns arrive through renegotiation with the incumbent rather than through job switching. Where Section 6 measured how much competition contributes to managerial pay, this section asks how firm-specific capital impacts the way it operates.

Because it is forfeited when a manager changes employers, firm-specific human capital raises the private cost of moving and creates *job lock*, reducing cross-firm mobility and changing how labor market competition affects managerial compensation. The portability counterfactual quantifies these effects along two margins. First, making capital portable lowers the share of CEO appointments filled from within from 68% to 64%, with reallocation going to long-tenured managers and those at large firms. Second, firm-specific human capital shifts contractual returns from job switching toward renegotiation with the incumbent. Together, these mobility and contracting frictions reduce manager welfare, particularly for managers late in their careers at a firm.

We isolate the role of firm-specific human capital by comparing the baseline economy with one in which it is *portable*. That is, $k(\tau)$ and its productive content are unchanged, but $k(\tau)$ no longer resets when a manager moves. The comparison therefore preserves the productive value of accumulated human capital while isolating the labor-market effects of its non-portability.

7.1. Human Capital and Transitions into CEO Employment

Table 6 reports the baseline and counterfactual composition of CEO appointments: filled from within the firm, by poaching executives, or by poaching sitting CEOs. Making human capital portable leads to a 4.4 pp reallocation toward external hiring, a 14% increase relative to the baseline external-hiring share, split between poached executives (+2.3 pp) and poached sitting CEOs (+2 pp). Non-portability therefore accounts for a modest part of firms' tilt toward internal hiring: shutting the friction off entirely moves the internal share from 68.1% to 63.7%, and internal appointment remains much the most common route to the job, consistent with the hiring patterns documented in [Cziraki and Jenter \(2022\)](#).²²

The reallocation is concentrated where accumulated capital is greatest, which is where job lock should bind. The external-hiring share is essentially unchanged among matches with fewer than five years of tenure but rises by 7 pp, or 26% of the baseline external-hiring rate, among those with ten to twenty-five years. The pattern repeats across firm size: portability raises external hiring by 3 pp at small firms and 5 pp at large firms, increases of 6% and 37% relative to baseline rates. Because large firms already have high productivity, fewer outside offers can clear the manager's incumbent match value, making accumulated firm-specific human capital especially binding.

The larger part of the tilt reflects an offer-arrival gap that the model takes as given. Our estimates imply $\lambda_0 > \lambda_1$, so insiders receive CEO offers more readily than outsiders, but we cannot determine whether this gap reflects, for example, an inefficient preference for internal candidates ([He and Schroth, 2024](#)) or the efficient avoidance of external search costs ([Geelen and Hajda, 2024](#)). Competition reinforces this: an external firm must contend with the incumbent, which can respond through contract renegotiation and reduce the probability of a successful poach, whereas internal promotion faces no such pressure. This retention channel also determines whether competitive gains arise through renegotiation or job switching, which we examine next.

²²Table B.2 reports the same counterfactual from the sitting executive's perspective, tabulating the rates at which executives are promoted to CEO internally, promoted to CEO at another firm, or move laterally to another firm. It records the mobility margin underlying the reallocation described here: making capital portable lowers the internal-promotion rate and raises the rate of horizontal moves, with the effect again concentrated among high-tenure executives and those at large firms.

7.2. Sources of Contractual Returns

Firm-specific human capital shapes a manager's contractual returns through two channels. On the extensive margin, by raising the productivity a poaching firm must offer to win her, it widens the range of outside offers that are met by renegotiation rather than by a move (Proposition 2(i)), making renegotiation a more common response to competition. On the intensive margin, the same accumulated capital raises the value of the incumbent match, so any given outside offer poses a weaker threat and can be countered with a smaller concession (Proposition 2(ii)). Managers therefore renegotiate more often but extract less each time, leaving the net effect ambiguous.

Firm-specific human capital changes the composition of contractual returns considerably more than their size. Cumulative career piece-rate growth is 0.759 log points in the baseline and 0.773 with portable human capital. The two margins therefore nearly offset: job lock reduces cumulative contractual returns by under two percent, with smaller concessions on the intensive margin slightly outweighing additional renegotiations on the extensive margin. What changes is where those returns come from. In the baseline economy, 72.8% of this growth is realized through renegotiation with the incumbent and 27.2% through cross-firm job switches; making capital portable shifts 4.9 percentage points from renegotiation to switching, a rise of 18 percent in the switching share. Portability therefore affects the composition of managers' contractual returns about ten times more than their total size.

The reallocation is concentrated where job lock binds hardest. Among managers with ten to twenty-five years of tenure, portability raises the switching share by 5.7 percentage points; at small firms it raises it by 11.3 points, from 34.5% to 45.8%, a relative increase of roughly a third. The total effect changes sign across the same cuts: job lock raises cumulative contractual returns for managers in their first ten years at a firm and lowers them thereafter. These comparisons also reflect portability's endogenous effects on mobility and selection, since the bins do not follow the same matches through both economies; the next subsection reports the corresponding welfare cuts matched within manager.

7.3. The Managerial Welfare Impacts of Firm-Specific Human Capital

We now ask what these effects imply for manager welfare, isolating the labor-market cost of non-portability while holding fixed the productive value of firm-specific capital. Any welfare difference therefore arises because non-portability changes whether outside offers lead to switching or renegotiation, how much surplus managers capture, where they work, and whether they are promoted; it excludes the direct productivity loss from destroyed human capital at a move.

Incumbent firms can partly offset this cost through renegotiation, raising compensation when outside offers arise and retaining managers who would otherwise leave; we measure the welfare cost that remains after these endogenous responses. We evaluate welfare using the contractual and allocative component of log wages:

$$v_{it} = r_{it} + f_{it} + \Delta_C \cdot \mathbf{1}\{\text{CEO}\}_{it} \quad (21)$$

which captures surplus extraction, firm allocation, and promotion to CEO.

Inspired by [Lucas \(1987\)](#), we express the difference in lifetime value as a *compensation equivalent*: the constant proportion by which flow compensation must be scaled to make a manager indifferent between the locked and portable-capital economies. Let $\Delta \bar{V}^v$ denote the difference in the expected discounted value of this component between the two economies and $\bar{H}$ the expected discounted time attached to a firm. Under log utility:

$$\text{CE} = \exp\left(\frac{\Delta \bar{V}^v}{\bar{H}}\right) - 1 \quad (22)$$

A positive (negative) CE means job lock raises (lowers) welfare. [Table 8](#) reports these values, and [Appendix C.8](#) provides the derivation and estimation details.

On average, job lock imposes a modest welfare cost: setting aside the capital a move destroys, the representative manager would accept a permanent 0.59% reduction in flow compensation to escape it. This cost reflects reduced surplus capture (−0.63%) and slower promotion into CEO

positions (-0.21%), partly offset by allocation to more productive firms ($+0.25\%$). The average masks a sharp reversal over the career, the welfare counterpart of the sign change in cumulative returns reported above: the CE is $+1.65\%$ at tenures below five years but -2.51% between ten and twenty-five years. At low tenure, the option value of incumbent renegotiation dominates because managers have accumulated too little firm-specific human capital to be meaningfully locked in. Surplus capture contributes $+1.08\%$ and allocation to more productive firms a further $+0.71\%$, against -0.13% from slower promotion. As tenure rises, the loss of mobility and bargaining leverage outweighs this benefit. A similar reversal appears across firm size: the CE is $+0.30\%$ at small firms but -1.19% at large firms. At large firms, accumulated human capital raises an already valuable incumbent match, weakening outside offers as a source of bargaining leverage.

More broadly, the portability counterfactual separates the productive and contractual roles of firm-specific human capital. The same capital that makes the incumbent match more valuable also weakens the manager's outside options, shifting competition away from mobility and reducing bargaining leverage. These costs are back-loaded, reaching 2.51% of compensation for managers ten to twenty-five years into a match. Thus, the productive attachment that makes a manager valuable to her firm increasingly limits the competition for her services.

8. Conclusion

We present a structural model of the managerial labor market that quantifies the roles of general and firm-specific human capital accumulation, managerial bargaining power, and labor market competition in shaping compensation and mobility. Our results show that limited managerial mobility need not imply weak competition: outside offers can substantially raise incumbent contracts without producing job switches. Bargaining power alone would give managers 31.6% of match surplus, while competition lifts realized capture to 54.6% , and to 79.3% for externally hired CEOs. High realized surplus capture thus need not reflect managerial bargaining power alone.

Firm-specific human capital is central to this relationship between competition and mobility. As firm-specific skill accumulates, the incumbent match becomes increasingly valuable relative

to outside opportunities. Competition consequently operates less through poaching and more through improved contracts at the incumbent firm. This mechanism also helps explain the predominance of internal CEO appointments, although differences in the arrival of internal and external opportunities remain quantitatively more important. Thus, firm-specific human capital changes not only managers' productivity and mobility, but also the form through which labor market competition affects their careers.

The non-portability of firm-specific human capital also has important welfare consequences. While productive attachment to the incumbent firm can benefit managers early in a match, accumulated firm-specific skill eventually restricts mobility and weakens outside bargaining leverage. More broadly, our results emphasize that the competitiveness of the managerial labor market cannot be inferred from observed mobility alone. Understanding managerial pay and career dynamics requires accounting jointly for the composition of managerial human capital, competition for talent, and bargaining within ongoing employment relationships.

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Figure 1. Cumulative wage-experience and wage-tenure profiles

This figure displays cumulative wage-experience and wage-tenure profiles. Panel A and Panel B compare the experience and tenure profiles based on the Mincer wage regression (16), by using the tenure and experience polynomial coefficients shown in Table 2, Panel B. Panel C plots the structurally-estimated tenure and experience accumulation profiles based using the tenure and experience cubic coefficients in Table 3.

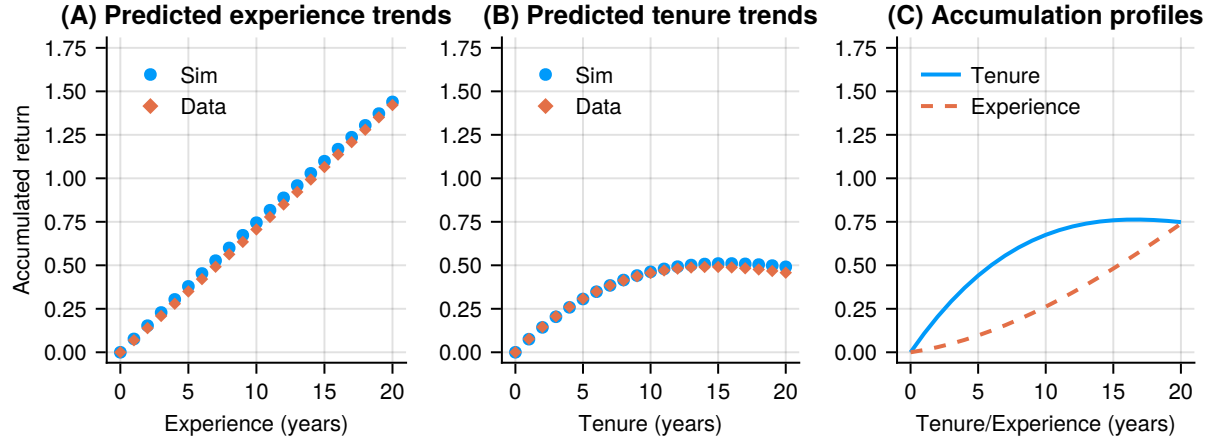


Figure 2. Compensation Dynamics around CEO Appointment: Data vs. Model

This figure plots event-time coefficients β_k from $\log w_{it} = \sum_{k=-5}^5 \beta_k \mathbf{1}\{t - T_i = k\} + \mu_{ij} + \lambda_t + \varepsilon_{it}$, estimated separately on the Execucomp panel (orange, with 95% confidence intervals clustered by manager) and on the simulated panel at the estimated parameters (blue), where T_i is the year of manager i 's first appointment as CEO, pooling internal promotions and external CEO hires; μ_{ij} is a manager $\times$ firm (match) fixed effect, and λ_t a year effect. β_{-1} is normalized to zero; event time beyond ± 5 is binned; managers never appointed CEO serve as controls. Because the match effect is specific to a manager-firm pair, it absorbs part of the level change at external appointments, which involve a change of firm, whereas internal promotions identify the within-firm change in pay. Data events are dated to the first full fiscal year as CEO, dropping the transition year whose compensation blends the two roles. The model is not estimated on event-time moments.

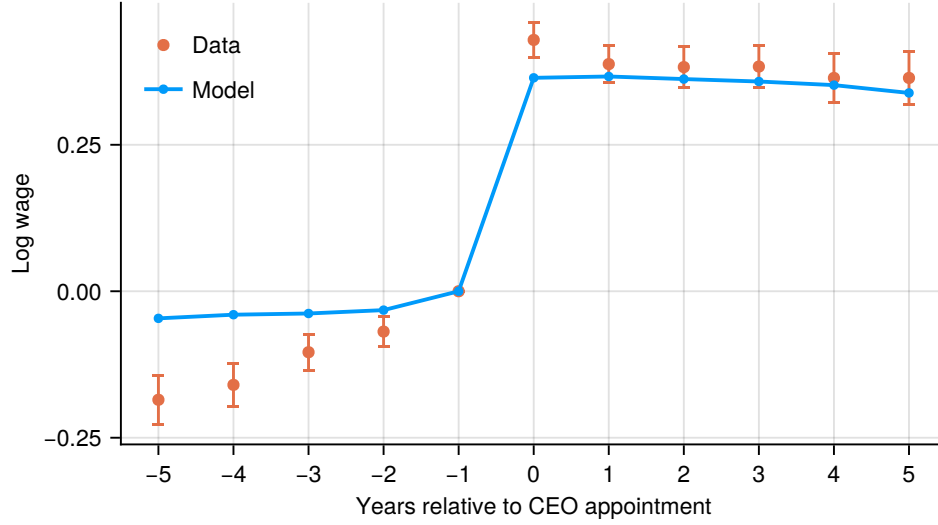


Figure 3. Contractual Returns Decomposition: Job Switches vs. Renegotiations

This figure displays a decomposition of cumulative piece rate growth across labor market experience and tenure. The solid (blue) line displays cumulative piece rate growth across labor market experience (tenure in Panel B). The dashed (orange) line displays contractual returns due to job-switches (i.e., accumulated piece rate growth due to switching firms). The dashed-dotted (green) line displays contractual returns due to renegotiations (i.e., accumulated piece rate growth due to renegotiations with the current firm).

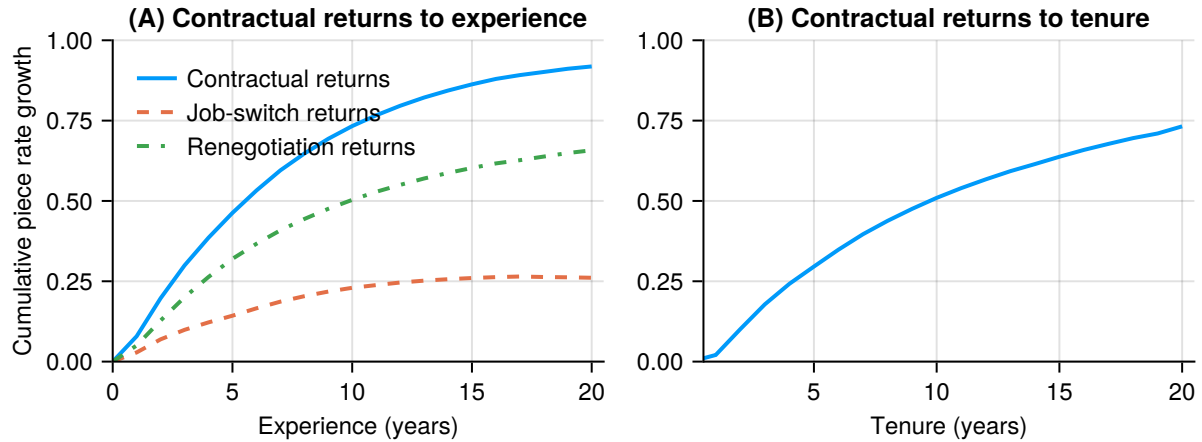


Table 1. Summary and labor market transition statistics

This table displays summary statistics for managerial wages, and labor market experience and tenure (Panel A); and labor market transition statistics (Panel B). In Panel A, statistics are displayed based on employment status (executive or CEO). In Panel B, labor market transition statistics are displayed conditional on employment state (executive, CEO, unattached); the first two columns display the number of transitions, and the proportion of observations conditional on employment state (i.e., the rate at which the transition occurs cross-sectionally within each group CEO, non-CEO executive, and unattached). The final two columns display the average years of experience and tenure that the transition occurs, respectively.

Panel A: Summary statistics

	N	Mean	Std Dev	25%	50%	75%
CEOs						
Log wage	61,558	14.857	1.208	14.067	14.917	15.685
Wage (\$ million)	61,558	5.342	10.872	1.286	3.007	6.485
Wage Growth (%)	56,245	9.772%	73.908%	-15.123%	7.432%	35.274%
Experience	61,558	22.186	11.719	13	21	31
Tenure	61,558	15.267	11.543	6	12	22
Non-CEO executives						
Log wage	262,162	13.790	1.067	13.051	13.773	14.507
Wage (\$ million)	262,162	1.744	3.905	0.465	0.958	1.997
Wage Growth (%)	210,196	13.734%	64.974%	-11.185%	9.573%	36.749%
Experience	262,162	15.239	11.901	5	12	23
Tenure	262,162	5.776	5.848	2	4	7

Panel B: Labor market transition statistics

	N	%	Experience at transition	Tenure at transition
CEOs				
Horizontal CEO move	293	0.476%	22.631	12.840
Unattachment	543	0.882%	20.722	11.560
Retirement	8,735	14.190%	23.673	15.538
Non-CEO executives				
Internal CEO promotion	4,170	1.591%	17.190	10.474
External CEO promotion	374	0.143%	16.321	5.495
Horizontal executive move	1,535	0.586%	14.383	4.805
Unattachment	5,098	1.945%	15.048	4.595
Retirement	41,169	15.704%	15.680	6.340
Unattached managers				
Executive position	5,085	22.644%	19.197	
CEO position	823	3.665%	22.176	

Table 2. Model fit

This table compares observed and simulated moments, given the data described in Section 2 and the estimation and identification strategies described in Section 4. “Observed” refers to the real data. “Simulated” refers to the simulated data. Standard errors are displayed below each moment in parentheses. Standard errors for the empirical moments are computed using an influence function approach and are clustered at the manager level. Standard errors for the simulated moments are computed using the standard GMM moment variance implied by the simulated data. Moments in Panel A concern labor market mobility and steady-state shares of employment. Moments in Panel B concern the two auxiliary regressions.

Panel A: Labor market mobility

	Description	Observed	Simulated	Diff
(1)	Unattachment shock (0–5 yrs)	0.059 (0.002)	0.048	-0.011
(2)	Unattachment shock (5–10 yrs)	0.051 (0.002)	0.056	0.005
(3)	Internal CEO hire (0–5 yrs)	0.151 (0.001)	0.148	-0.003
(4)	Internal CEO hire (5–10 yrs)	0.145 (0.002)	0.127	-0.018
(5)	External CEO hire (0–5 yrs)	0.016 (0.001)	0.016	-0.000
(6)	External CEO hire (5–10 yrs)	0.014 (0.001)	0.013	-0.001
(7)	External exec hire (0–5 yrs)	0.083 (0.001)	0.074	-0.009
(8)	External exec hire (5–10 yrs)	0.054 (0.002)	0.058	0.004
(9)	Unattachment to exec (0–2 yrs)	0.412 (0.008)	0.430	0.018
(10)	Unattachment to CEO (0–2 yrs)	0.065 (0.019)	0.045	-0.020
(11)	CEO share	0.343 (0.003)	0.372	0.029

Panel B: Mincer regression

	Description	Observed	Simulated	Diff
(1)	Firm effect (mean)	12.508 (0.014)	12.518	0.010
(2)	Firm effect (variance)	0.290 (0.024)	0.275	-0.015
(3)	Firm effect (skewness)	-0.117 (0.349)	0.389	0.506
(4)	Firm effect (size slope)	0.129 (0.007)	0.128	-0.001
(5)	Manager effect (size slope)	0.156 (0.007)	0.151	-0.005
(6)	Manager effect (variance)	1.528 (0.019)	1.532	0.004
(7)	Tenure (linear)	0.081 (0.003)	0.079	-0.002
(8)	Tenure (quadratic)	-0.412 (0.021)	-0.383	0.029
(9)	Tenure (cubic)	0.616 (0.038)	0.554	-0.062
(10)	Experience (linear)	0.069 (0.003)	0.077	0.008
(11)	Experience (quadratic)	0.023 (0.014)	-0.025	-0.048
(12)	Experience (cubic)	-0.061 (0.021)	-0.001	0.060
(13)	CEO indicator	0.568 (0.011)	0.522	-0.046
(14)	Wage residual (variance)	0.305 (0.008)	0.316	0.011
(15)	Wage residual (autocovariance)	0.355 (0.035)	0.322	-0.033

Panel C: Wage growth regression

	Description	Observed	Simulated	Diff
(1)	Tenure growth (quadratic)	-0.366 (0.032)	-0.136	0.230
(2)	Tenure growth (cubic)	0.556 (0.055)	0.260	-0.296
(3)	Experience growth (quadratic)	-0.205 (0.022)	-0.219	-0.014
(4)	Experience growth (cubic)	0.182 (0.033)	0.213	0.031
(5)	CEO growth	0.404 (0.012)	0.469	0.065
(6)	Wage growth residual (variance)	0.398 (0.013)	0.435	0.037
(7)	Wage growth residual (autocovariance)	-0.377 (0.020)	-0.448	-0.071

Table 3. Structural parameter estimates

This table displays the estimated structural parameters. Parameter standard errors, derived via the Delta method using the optimal weight matrix, are displayed in parentheses next to each parameter.

Description	Notation	Parameter value	
Arrival rates			
Internal CEO offer	λ_0	0.0295	(0.0007)
External CEO offer	λ_1	0.0070	(0.0004)
External exec offer	λ_2	0.0368	(0.0013)
Unattachment shock	η	0.0220	(0.0012)
Unattachment to exec	γ_E	0.2496	(0.0060)
Unattachment to CEO	γ_C	0.0221	(0.0023)
Bargaining Power			
Managerial bargaining power	β	0.3164	(0.0415)
Productivity and sorting			
Location	f_{min}	12.3809	(0.1808)
Weibull scale	s_1	1.9229	(0.1968)
Weibull shape	s_2	3.1861	(0.9454)
CEO productivity premium	δ_C	0.5992	(0.1009)
Passthrough of firm size to productivity	b_{size}	0.1818	(0.0354)
Manager-firm sorting	ρ_{sort}	0.4846	(0.0259)
Manager ability (volatility)	σ_a	1.6775	(0.0125)
Measurement error			
Idiosyncratic shock (volatility)	σ_e	0.6702	(0.0053)
Idiosyncratic shock (persistence)	ν_e	0.0184	(0.0476)
Human capital accumulation			
Tenure polynomial (linear)	c_1^k	0.1125	(0.0040)
Tenure polynomial (quadratic)	$c_2^k \times 10^2$	-0.5243	(0.0222)
Tenure polynomial (cubic)	$c_3^k \times 10^4$	0.7445	(0.0395)
Experience polynomial (linear)	c_1^g	0.0115	(0.0048)
Experience polynomial (quadratic)	$c_2^g \times 10^2$	0.1676	(0.0201)
Experience polynomial (cubic)	$c_3^g \times 10^4$	-0.2010	(0.0270)

Table 4. Dispersion in managerial compensation

Panel A decomposes total wage variance $\text{Var}(w)$. Panel B conditions on a and decomposes the within-ability variance $\text{Var}(w - a)$, which equals $\approx 47.8\%$ of total variance; each Panel B entry is the corresponding Panel A entry divided by that factor.

Panel A: Unconditional (share of $\text{Var}(w)$)

	(1) Full sample	(2) Conditional on experience (t)	(3)	(4)	(5) Conditional on tenure (τ)	(6)	(7)
		[0, 5)	[5, 10)	[10, 25)	[0, 5)	[5, 10)	[10, 25)
$\text{Var}(r)$	18.18%	23.20%	22.13%	16.70%	22.75%	19.82%	16.68%
$\text{Var}(f)$	14.91%	21.16%	17.76%	16.42%	17.13%	16.70%	16.58%
$\text{Var}(g)$	2.81%	0.03%	0.08%	1.77%	1.79%	1.76%	1.96%
$\text{Var}(k)$	1.82%	0.61%	1.07%	1.36%	0.55%	0.19%	0.03%
$\text{Var}(a)$	46.69%	60.12%	53.81%	54.52%	51.53%	51.39%	55.00%
$\text{Var}(u)$	6.94%	9.30%	8.04%	8.00%	7.69%	7.61%	8.17%
$2 \text{Cov}(r, f)$	-7.73%	-21.53%	-10.30%	-7.76%	-10.34%	-7.80%	-8.73%
$2 \text{Cov}(r, g)$	4.33%	0.32%	0.31%	1.10%	4.23%	2.90%	1.61%
$2 \text{Cov}(r, k)$	2.19%	0.13%	-1.08%	-0.49%	-0.06%	0.27%	0.07%
$2 \text{Cov}(r, a)$	-4.04%	-11.96%	-4.66%	-2.33%	-7.40%	-4.38%	-2.45%
$2 \text{Cov}(f, g)$	1.34%	0.01%	0.09%	0.54%	1.45%	1.33%	0.67%
$2 \text{Cov}(f, k)$	0.28%	-0.31%	-0.52%	-0.52%	-0.26%	0.09%	0.02%
$2 \text{Cov}(f, a)$	9.72%	18.75%	12.23%	9.60%	11.32%	10.60%	10.85%
$2 \text{Cov}(g, k)$	2.72%	0.18%	0.19%	0.66%	0.19%	0.25%	0.20%
$2 \text{Cov}(g, a)$	-0.35%	-0.09%	-0.02%	-0.06%	-0.57%	-0.73%	-0.65%
$2 \text{Cov}(k, a)$	0.20%	0.05%	0.82%	0.52%	-0.02%	0.01%	0.01%

Panel B: Conditional on ability a

	(1) Full sample	(2) Conditional on experience (t)	(3)	(4)	(5) Conditional on tenure (τ)	(6)	(7)
		[0, 5)	[5, 10)	[10, 25)	[0, 5)	[5, 10)	[10, 25)
Var(r)	38.03%	70.25%	58.53%	44.20%	50.42%	45.96%	44.76%
Var(f)	31.20%	64.07%	46.96%	43.44%	37.96%	38.73%	44.49%
Var(g)	5.88%	0.08%	0.21%	4.69%	3.96%	4.07%	5.27%
Var(k)	3.81%	1.86%	2.84%	3.60%	1.21%	0.45%	0.09%
Var(u)	14.52%	28.15%	21.26%	21.18%	17.05%	17.64%	21.93%
2 Cov(r, f)	-16.17%	-65.19%	-27.23%	-20.55%	-22.91%	-18.08%	-23.44%
2 Cov(r, g)	9.05%	0.96%	0.81%	2.90%	9.38%	6.72%	4.32%
2 Cov(r, k)	4.59%	0.38%	-2.86%	-1.30%	-0.14%	0.62%	0.20%
2 Cov(f, g)	2.80%	0.02%	0.23%	1.42%	3.21%	3.09%	1.80%
2 Cov(f, k)	0.59%	-0.94%	-1.38%	-1.38%	-0.57%	0.21%	0.06%
2 Cov(g, k)	5.69%	0.55%	0.51%	1.76%	0.42%	0.57%	0.52%

Table 5. Managerial surplus capture decomposition

This table shows realized managerial surplus capture. Panel A displays realized surplus capture by CEO hire type. Panel B displays results for the full managerial sample and across labor market experience, firm tenure, and firm size. In each cut, we report the average realized surplus capture $\bar{b}$ and the share attributable to labor market competition, $\left(\frac{\bar{b}-\beta}{\bar{b}}\right)$, where $\beta = 31.6\%$ is the ex ante bargaining share. In Panel A, we focus exclusively on CEOs in the simulated sample, so the averages in column 1 refer only to CEOs. In Panel B, we focus on the full sample and include executives in the analysis.

Panel A: Surplus capture by CEO hire type

	(1)	(2)
	Realized surplus capture ($\bar{b}$)	Labor market competition $\left(\frac{\bar{b}-\beta}{\bar{b}}\right)$
Full CEO sample	54.26%	41.68%
Internal CEO	49.62%	36.23%
External CEO	79.28%	60.09%

Panel B: Surplus capture across the managerial labor market

	(1)	(2)
	Realized surplus capture ($\bar{b}$)	Labor market competition $\left(\frac{\bar{b}-\beta}{\bar{b}}\right)$
Full Sample	54.59%	42.03%
By Tenure		
$\tau \in [0, 5)$	64.83%	51.19%
$\tau \in [5, 10)$	55.44%	42.93%
$\tau \in [10, 25)$	51.32%	38.34%
By Experience		
$t \in [0, 5)$	53.02%	40.32%
$t \in [5, 10)$	54.51%	41.96%
$t \in [10, 25)$	54.59%	42.03%
By Firm Size		
Small	48.16%	34.30%
Medium	55.54%	43.03%
Large	59.85%	47.13%

Table 6. Firm-Specific Human Capital and the Path to CEO

This table reports how CEO positions are filled in the baseline economy (firm-specific k) and in the counterfactual with no k -lock (portable k); the Diff columns report no-lock – baseline (in percentage points). Internal = internal promotion, Poached Executive = external hire of a non-CEO executive, and Poached CEO = external hire of a sitting CEO. Rows report the full sample and cuts by firm tenure, labor market experience, and firm size. Table B.2 reports the corresponding transition rates from the sitting executive's perspective.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Internal			Poached Executive			Poached CEO		
	Base	No k -lock	Diff	Base	No k -lock	Diff	Base	No k -lock	Diff
Full Sample	68.07%	63.69%	-4.38 p.p.	24.77%	27.11%	+2.34 p.p.	7.17%	9.20%	+2.04 p.p.
By Tenure									
$\tau \in [0, 5)$	62.32%	62.45%	+0.13 p.p.	32.74%	32.16%	-0.57 p.p.	4.94%	5.39%	+0.44 p.p.
$\tau \in [5, 10)$	67.39%	62.43%	-4.95 p.p.	24.43%	28.50%	+4.06 p.p.	8.18%	9.07%	+0.89 p.p.
$\tau \in [10, 25)$	73.26%	66.29%	-6.98 p.p.	18.11%	21.99%	+3.89 p.p.	8.63%	11.72%	+3.09 p.p.
By Experience									
$t \in [0, 5)$	53.08%	51.37%	-1.71 p.p.	41.10%	42.35%	+1.25 p.p.	5.82%	6.28%	+0.46 p.p.
$t \in [5, 10)$	62.65%	58.94%	-3.72 p.p.	29.14%	32.38%	+3.25 p.p.	8.21%	8.68%	+0.47 p.p.
$t \in [10, 25)$	73.19%	67.77%	-5.43 p.p.	19.42%	22.49%	+3.07 p.p.	7.39%	9.74%	+2.35 p.p.
By Firm Size									
Small	48.83%	45.88%	-2.95 p.p.	35.08%	35.84%	+0.76 p.p.	16.09%	18.28%	+2.19 p.p.
Medium	68.69%	63.04%	-5.65 p.p.	26.54%	29.22%	+2.68 p.p.	4.77%	7.74%	+2.97 p.p.
Large	86.65%	81.67%	-4.97 p.p.	12.70%	16.48%	+3.78 p.p.	0.65%	1.85%	+1.20 p.p.

Table 7. Firm-Specific Human Capital and the Source of Contractual Returns

This table compares the size and composition of managers' cumulative contractual returns in the baseline economy (firm-specific k) and in the counterfactual with no k -lock (portable k); the Diff columns report no-lock – baseline (Δr in log points, shares in percentage points). Columns 1–3 report the total cumulative career piece-rate change (Δr , in log points). Columns 4–9 decompose that total into the share captured through job-switching (across firms) and the share captured through renegotiation with the incumbent (within firm, including internal promotion), which sum to 100%. The sample is market-exposed executives; rows report the full sample and cuts by role, firm tenure, labor market experience, and firm size.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Total (Δr)			Job-switch (%)			Renegotiation (%)		
	Base	No k -lock	Diff	Base	No k -lock	Diff	Base	No k -lock	Diff
Full Sample	0.759	0.773	+0.014	27.18%	32.07%	+4.89 p.p.	72.82%	67.93%	-4.89 p.p.
By Role									
Executive	0.903	0.920	+0.017	25.27%	30.86%	+5.59 p.p.	74.73%	69.14%	-5.59 p.p.
CEO	0.499	0.509	+0.010	33.39%	35.98%	+2.59 p.p.	66.61%	64.02%	-2.59 p.p.
By Tenure									
$\tau \in [0, 5)$	0.821	0.784	-0.037	55.46%	52.99%	-2.47 p.p.	44.54%	47.01%	+2.47 p.p.
$\tau \in [5, 10)$	0.790	0.760	-0.030	36.86%	37.40%	+0.54 p.p.	63.14%	62.60%	-0.54 p.p.
$\tau \in [10, 25)$	0.769	0.787	+0.018	23.94%	29.64%	+5.70 p.p.	76.06%	70.36%	-5.70 p.p.
By Experience									
$t \in [0, 5)$	0.995	0.992	-0.003	34.65%	33.75%	-0.90 p.p.	65.35%	66.25%	+0.90 p.p.
$t \in [5, 10)$	0.935	0.935	+0.000	32.03%	34.63%	+2.60 p.p.	67.97%	65.37%	-2.60 p.p.
$t \in [10, 25)$	0.809	0.818	+0.009	28.26%	33.46%	+5.20 p.p.	71.74%	66.54%	-5.20 p.p.
By Firm Size									
Small	0.460	0.422	-0.038	34.49%	45.82%	+11.33 p.p.	65.51%	54.18%	-11.33 p.p.
Medium	0.795	0.811	+0.016	32.82%	38.68%	+5.86 p.p.	67.18%	61.32%	-5.86 p.p.
Large	1.021	1.087	+0.066	19.50%	21.80%	+2.30 p.p.	80.50%	78.20%	-2.30 p.p.

Table 8. Welfare Effects of Job Lock from Firm-Specific Human Capital

Each entry is a compensation-equivalent (CE): the constant percentage by which a manager's flow compensation in the portable- k world (same productivity, no job lock) would have to be permanently scaled to leave her exactly as well off as under job lock, matched within manager over the manager's expected future career. A positive CE means job lock raises welfare. Values are net of k 's output level, so they price the labor-market friction rather than the productivity of human capital: the CE converts the net-of-level lifetime-value gain from the lock, $\Delta \bar{V}^v$, into compensation units via

$$\text{CE} = \exp(\Delta \bar{V}^v / \bar{H}) - 1,$$

where $\bar{H}$ is the expected discounted attached lifetime and $\Delta \bar{V}^v = \bar{V}_{\text{locked}}^v - \bar{V}_{\text{lock-off}}^v$ is the change in the net-of-level value $v = r + f + \Delta_C \mathbf{1}\{\text{CEO}\}$. The Surplus Capture (r), Firm Productivity (f), and CEO Premium (Δ_C) columns apportion the Total in proportion to each channel's share of $\Delta \bar{V}^v$ and sum to it. Rows report the full sample and cuts by role, firm tenure, labor market experience, and firm size. Appendix C.8 provides details on the construction of the compensation-equivalence.

	(1)	(2)	(3)	(4)
	Total	Surplus Capture (r)	Firm Productivity (f)	CEO Premium (Δ_C)
Full Sample	-0.59%	-0.63%	+0.25%	-0.21%
By Role				
Executive	-0.55%	-0.36%	+0.14%	-0.32%
CEO	-0.69%	-1.28%	+0.52%	+0.07%
By Tenure				
$\tau \in [0, 5)$	+1.65%	+1.08%	+0.71%	-0.13%
$\tau \in [5, 10)$	-0.05%	-0.14%	+0.30%	-0.20%
$\tau \in [10, 25)$	-2.51%	-2.14%	-0.09%	-0.28%
By Experience				
$t \in [0, 5)$	+0.80%	-0.29%	+1.20%	-0.11%
$t \in [5, 10)$	-0.31%	-0.40%	+0.26%	-0.17%
$t \in [10, 25)$	-1.14%	-0.79%	-0.07%	-0.27%
By Firm Size				
Small	+0.30%	+0.47%	-0.03%	-0.15%
Medium	-0.87%	-0.87%	+0.30%	-0.30%
Large	-1.19%	-1.48%	+0.47%	-0.18%

A. Model Appendix

A.1. Preliminaries

Notation follows the main text. In addition, we let $\bar{\Psi}(\cdot \mid a) = 1 - \Psi(\cdot \mid a)$ denote the survivor function of the offer distribution and define $\gamma \equiv \delta + \mu + \eta$ for brevity. Recall that $V_i(r, \tau, f)$ denotes the value of employment in a type- i role, $i \in \{E, C\}$, and that the value of unattachment satisfies $V_U = V_E(0, 0, f_{\min})$.

The main text suppresses experience from the state and treats ability as affecting decisions only through the offer distribution. The following lemma justifies both steps.

Lemma 1. *Suppose all arrival rates, δ , μ , and $\Psi(\cdot \mid a)$ are independent of t , general capital $g(t)$ accrues in every labor-market state and is retained on every transition, and ability a is permanent. Then every value function decomposes as:*

$$V_i(r, t, \tau, f; a) = A(t) + \frac{a}{\delta + \mu} + B_i(r, \tau, f; a), \quad A(t) \equiv \int_0^\infty e^{-(\delta + \mu)s} g(t + s) ds \quad (\text{A.1})$$

where the functions B_i solve a system identical to the original one with flow payoff $r + k(\tau) + f + \Delta_C \mathbf{1}\{i = C\}$, and a enters that system only through $\Psi(\cdot \mid a)$. Consequently, every bargaining rule, threshold, and acceptance decision is independent of t , and independent of a conditional on the offer distribution.

Proof. Guess the form (A.1) and substitute into the system of Section A.3. The flow payoff in every state is $g(t) + a$ plus a component free of t and a ; every transition preserves t and a and continues the accrual of g ; and retirement, which terminates all flows, is absorbed into the effective discount rate $\delta + \mu$. Differentiating A yields:

$$A'(t) = (\delta + \mu)A(t) - g(t) \quad (\text{A.2})$$

so the terms $A(t) + a/(\delta + \mu)$ satisfy $(\delta + \mu)A = g + A'$ and $(\delta + \mu) \cdot a/(\delta + \mu) = a$ state by state, and cancel from both sides of every equation in the system, leaving the stated t -free system for the

B_i . The unattached value inherits the decomposition through the condition $V_U = V_E(0, 0, f_{min})$, which holds at every (t, a) . Finally, every sharing rule and threshold condition below is either an affine relation among contemporaneous values of the same manager whose coefficients sum to one, or an equality between two such values. Adding the common constant $A(t) + a/(\delta + \mu)$ to every value therefore leaves each relation unchanged. ■

In the remainder of the appendix we accordingly write $V_i(r, \tau, f)$ for the component B_i , suppressing t entirely and suppressing a except where the offer distribution is concerned. When the offer distribution is ability-conditional, the system is solved separately for each ability level, and all statements below hold ability by ability. Throughout Section A.2, the value functions are treated as given functions of the state, assumed only to be continuous and strictly increasing in both r and f . Section A.3 derives these functions and verifies both properties, so that the bargaining outcomes and the value functions are mutually consistent.

A.2. Bargaining

A.2.1. Contracts and the Sharing Rule

A contract is a piece rate r , binding until revised at a bargaining event. With compensation expense $W = RY$ where $R = e^r$, the firm's instantaneous profit flow from the match is:

$$Y - W = Y(1 - e^r) \geq 0 \iff r \leq 0 \tag{A.3}$$

for every realization of output. Any contract with $r \leq 0$ is feasible for the firm, and $r = 0$ (full extraction by the manager) is the boundary of firm participation: at $r = 0$ the firm earns zero profit at every instant. Therefore, the most a type- (i, τ, f) employer can credibly promise its manager is the full-extraction value $V_i(0, \tau, f)$. It is also useful to define:

$$\Pi_i(r, \tau, f) \equiv V_i(0, \tau, f) - V_i(r, \tau, f) \geq 0 \tag{A.4}$$

which we refer to as the *withheld value* of the match: the portion of the maximal deliverable value not conceded to the manager under the current contract. It is measured in units of the manager's lifetime value and serves only to record disagreement positions inside negotiations.

Every negotiation in the model is resolved by a common sharing rule. Consider a negotiation between the manager and a single firm, where the firm can credibly deliver the manager at most $\bar{V}$, the manager's disagreement value is T_m , and the firm's disagreement position is worth T_f in manager-value units. The disagreement position T_f is the withheld value of whatever match the firm retains if negotiations fail; a firm left without the manager retains no match, and we assign zero value to its vacant position, as in [Cahuc et al. \(2006\)](#), so that $T_f = 0$ in that case. If $\bar{V} \geq T_m + T_f$, agreement is reached and the contract delivers the manager the value:

$$V = T_m + \beta(\bar{V} - T_m - T_f) \quad (\text{A.5})$$

with $\beta \in [0, 1)$ the manager's bargaining power; if $\bar{V} < T_m + T_f$, no agreement is reached and each party retains its disagreement position. The rule awards the manager her disagreement value plus a share β of the gain the negotiation creates, measured in manager-value units. Following [Bagger et al. \(2014\)](#), we take (A.5) as given rather than deriving it from an explicit extensive-form bargaining game.

A.2.2. Between-Firm Competition

We now derive the outcomes of cross-firm offers summarized in Proposition 1 of the main text. The bargaining protocol follows [Cahuc et al. \(2006\)](#), with the negotiation outcomes given by (A.5). Consider a type- i manager in state (r, τ, f) who receives an offer for a type- j position from an outside firm of productivity f' . The game proceeds in three stages:

1. Both firms simultaneously make piece-rate offers: the incumbent offers a (possibly revised) contract in the type- i role at f , and the outside firm offers a contract in the type- j role at f' .

2. The manager either accepts one of the offers or retains her current contract.
3. If an offer was accepted at Stage 2, the manager may subsequently initiate a renegotiation with the firm whose offer was refused. This renegotiation is resolved by (A.5), with the manager's disagreement value equal to the value of the offer accepted at Stage 2 and $T_f = 0$, as the refused firm is left without the manager.

Because a cross-firm move resets tenure and destroys $k(\tau)$, the outside firm's full-extraction value is $V_j(0, 0, f')$, while the incumbent's is $V_i(0, \tau, f)$.

Proposition A.1 (restated from the main text). *Suppose a type- i manager with state (r, τ, f) receives an outside offer for a type- j position from a firm of productivity f' . There exist thresholds $\underline{\theta}_{ij}(r, \tau, f) \leq \bar{\theta}_{ij}(\tau, f)$ such that: (i) if $f' > \bar{\theta}_{ij}$, the manager leaves for the poaching firm; (ii) if $\underline{\theta}_{ij} < f' \leq \bar{\theta}_{ij}$, the incumbent retains the manager at a renegotiated contract; and (iii) if $f' \leq \underline{\theta}_{ij}$, the offer is ignored and the manager's contract is unchanged. The acceptance threshold satisfies:*

$$V_j(0, 0, \bar{\theta}_{ij}(\tau, f)) = V_i(0, \tau, f) \quad (\text{A.6})$$

and the renegotiation threshold satisfies:

$$V_i(r, \tau, f) = V_j(0, 0, \underline{\theta}_{ij}) + \beta \left(V_i(0, \tau, f) - V_j(0, 0, \underline{\theta}_{ij}) \right) \quad (\text{A.7})$$

Moreover, the acceptance threshold $\bar{\theta}_{ij}$ is increasing in the current firm's productivity f .

Proof. The game is solved by backward induction. At Stage 3, a manager who accepted an offer worth T at Stage 2 and renegotiates with the refused firm, whose full-extraction value is $\bar{V}$, obtains $T + \beta(\bar{V} - T)$ if $\bar{V} \geq T$ and retains the Stage-2 outcome otherwise.

Moving back to Stage 1, both firms make simultaneous offers. Suppose first that $V_j(0, 0, f') > V_i(0, \tau, f)$, so that the outside firm can deliver values the incumbent cannot match. Were the manager to accept the incumbent's best offer at Stage 2, worth at most $V_i(0, \tau, f)$ by (A.3), she

would subsequently renegotiate with the outside firm at Stage 3 and obtain:

$$V_j(r', 0, f') = V_i(0, \tau, f) + \beta \left(V_j(0, 0, f') - V_i(0, \tau, f) \right) \quad (\text{A.8})$$

The outside firm therefore cannot win the manager with any offer worth less than (A.8), while offering more would concede value unnecessarily; it accordingly bids r' satisfying (A.8) at Stage 1, which the manager accepts, and the Stage-3 renegotiation is left unexercised in equilibrium. The bid is feasible because, with $\beta < 1$, the value in (A.8) lies strictly below the outside firm's full-extraction value. The incumbent cannot afford to outbid: retaining the manager would require delivering more than $V_i(0, \tau, f)$, violating (A.3). Hence the manager moves at contract (A.8).

Suppose instead that $V_j(0, 0, f') \leq V_i(0, \tau, f)$, so that the incumbent can match any feasible outside offer and retains the manager. The manager's leverage is the threat of accepting the outside firm's maximal offer, worth $V_j(0, 0, f')$, and renegotiating with the incumbent at Stage 3, which by (A.5) would leave her with:

$$W_{ij}(f') \equiv V_j(0, 0, f') + \beta \left(V_i(0, \tau, f) - V_j(0, 0, f') \right) = (1 - \beta) V_j(0, 0, f') + \beta V_i(0, \tau, f) \quad (\text{A.9})$$

If $W_{ij}(f') > V_i(r, \tau, f)$, the incumbent forestalls this outcome by offering the revised contract r' satisfying:

$$V_i(r', \tau, f) = V_j(0, 0, f') + \beta \left(V_i(0, \tau, f) - V_j(0, 0, f') \right) \quad (\text{A.10})$$

which the manager rationally accepts. If $W_{ij}(f') \leq V_i(r, \tau, f)$, the outside offer generates no credible improvement, and a competing offer does not necessitate an alteration of the current piece rate: the offer is ignored. Note that $W_{ij}(f') \geq V_j(0, 0, f')$ in this case, so accepting the outside offer outright is also dominated.

It remains to express these cases as thresholds in f' and establish their properties. Because $x \mapsto V_j(0, 0, x)$ is continuous and strictly increasing, equation (A.6) has a unique solution $\bar{\theta}_{ij}(\tau, f)$ whenever $V_i(0, \tau, f)$ lies in its range, and the two cases above are exactly $f' > \bar{\theta}_{ij}$ and $f' \leq$

$\bar{\theta}_{ij}$; if $V_i(0, \tau, f)$ lies above (below) the range, no poacher can (every poacher can) deliver the incumbent's value, and $\bar{\theta}_{ij}$ is set to the corresponding endpoint of the support, with the region beyond it empty. Similarly, W_{ij} is continuous and strictly increasing in f' since $\beta < 1$, and (A.7) is precisely $W_{ij}(\underline{\theta}_{ij}) = V_i(r, \tau, f)$, so the renegotiation threshold is unique whenever it exists. For the ordering, evaluate W_{ij} at the acceptance threshold using (A.6):

$$W_{ij}(\bar{\theta}_{ij}) = (1 - \beta)V_i(0, \tau, f) + \beta V_i(0, \tau, f) = V_i(0, \tau, f) \geq V_i(r, \tau, f) = W_{ij}(\underline{\theta}_{ij}) \quad (\text{A.11})$$

where the inequality holds because V_i is strictly increasing in r and $r \leq 0$. Since W_{ij} is strictly increasing, $\underline{\theta}_{ij} \leq \bar{\theta}_{ij}$, with equality if and only if $r = 0$. Finally, by (A.6):

$$\bar{\theta}_{ij}(\tau, f) = \left(V_j(0, 0, \cdot) \right)^{-1} \left(V_i(0, \tau, f) \right) \quad (\text{A.12})$$

The inverse of a continuous, strictly increasing function is strictly increasing, and $V_i(0, \tau, f)$ is strictly increasing in f , so $\bar{\theta}_{ij}$ is strictly increasing in f as a composition of strictly increasing functions. ■

The equilibrium is supported by the manager's option to accept the losing firm's best offer and renegotiate with the winner, a maneuver never exercised in equilibrium. This is why the manager's disagreement value in (A.8) and (A.10) is the losing firm's full-extraction value rather than her current contract: competition between the two firms, not the manager's own bargaining power, drives the losing firm's bid to its participation boundary $r = 0$. Note that this argument relies only on the participation constraint (A.3) and the ranking of offers by manager value, not on the specific form of the bilateral sharing rule.

A.2.3. Internal Promotion and the Firm's Outside Option

Internal promotion is a bilateral negotiation, with no third bidder, so (A.5) applies directly. The one difference from the cross-firm case is that the firm's disagreement position is not empty-handed: if the negotiation fails, the manager remains an executive at her current contract, and

the firm retains the executive match at that contract. The parties' disagreement positions are therefore:

$$T_m = V_E(r, \tau, f), \quad T_f = \Pi_E(r, \tau, f) = V_E(0, \tau, f) - V_E(r, \tau, f) \quad (\text{A.13})$$

and the maximal value the firm can deliver in the CEO role is $\bar{V} = V_C(0, \tau, f)$. The gain created by the negotiation is:

$$\bar{V} - T_m - T_f = V_C(0, \tau, f) - V_E(0, \tau, f) \equiv S(\tau, f) \quad (\text{A.14})$$

the joint surplus generated by the change in role. Applying (A.5), the promotion is executed if and only if:

$$S(\tau, f) \geq 0 \quad (\text{A.15})$$

and, when executed, the contract r' satisfies:

$$V_C(r', \tau, f) = V_E(r, \tau, f) + \beta S(\tau, f) \quad (\text{A.16})$$

which are the acceptance condition and sharing rule of the main text. Note that the manager's gain from promotion, $\beta S(\tau, f)$, is independent of her current piece rate: the contract r raises her disagreement value and the firm's withheld value one for one. The comparison with Section A.2.2 is the formal content of the observation that managers are in more favorable bargaining positions when fielding external offers: in a cross-firm negotiation, competition pushes the manager's disagreement value to the losing firm's full-extraction value and the winning firm's disagreement position to zero, whereas in an internal negotiation the firm's fallback retains the executive match and the manager's fallback is her current contract.

We next consider the enrichment of the firm's outside option described in footnote 9 of the main text, in which the firm can respond to a breakdown by filling the CEO position with a next-best internal candidate rather than retaining the incumbent as an executive. Suppose that installing the alternative candidate secures the firm a fraction $\phi \in [0, 1)$ of the promotion surplus,

with $\phi \rightarrow 1$ as the alternative approaches a perfect substitute for the focal manager, so that the firm's disagreement position becomes:

$$T_f = \Pi_E(r, \tau, f) + \phi S(\tau, f) \quad (\text{A.17})$$

The negotiation's gain becomes $(1 - \phi)S(\tau, f)$, which is nonnegative exactly when $S(\tau, f) \geq 0$, so the acceptance condition (A.15) is unchanged. Applying (A.5), the executed contract satisfies:

$$V_C(r', \tau, f) = V_E(r, \tau, f) + \beta(1 - \phi) S(\tau, f) \quad (\text{A.18})$$

Comparing (A.18) with (A.16), the two environments deliver identical contracts and identical promotion decisions once β is replaced by $\beta(1 - \phi)$. Because ϕ and β enter the manager's problem only through their product, they cannot be separately identified from data on contracts or transitions. We accordingly normalize $\phi = 0$ and interpret the estimated β as the manager's effective bargaining share, net of the firm's ability to promote from within.

Lastly, the contract of a manager exiting unattachment is (A.5) with $T_m = V_U$, $T_f = 0$, and $\bar{V} = V_j(0, 0, f')$:

$$V_j(r', 0, f') = \beta V_j(0, 0, f') + (1 - \beta) V_U \quad (\text{A.19})$$

An unattached manager accepts any executive offer, since $V_E(0, 0, f') \geq V_E(0, 0, f_{\min}) = V_U$ by monotonicity in f , and accepts a CEO offer if and only if $V_C(0, 0, f') \geq V_U$.

A.3. Value Functions

We now derive the value functions associated with CEO and executive employment. The value functions and the bargaining outcomes of Section A.2 are jointly determined: the equations below embed the bargaining outcomes in their option-value terms, while the thresholds and contracts of Section A.2 are functions of the value functions. An equilibrium of the model is a solution of this fixed point, and at the end of this section we verify that the solution possesses the monotonicity and continuity properties on which Section A.2 relies. Throughout, we write $\tilde{w}_i(r, \tau, f) = r +$

$k(\tau) + f + \Delta_C \mathbf{1}\{i = C\}$ for the flow payoff net of the separable components of Lemma 1.

A.4. CEO Value Function

By Proposition 1, the manager's value jumps to $(1 - \beta)V_C(0, \tau, f) + \beta V_C(0, 0, x)$ upon accepting an offer from a firm of productivity $x > \bar{\theta}_{CC}$, and to $(1 - \beta)V_C(0, 0, x) + \beta V_C(0, \tau, f)$ upon a renegotiation triggered by $x \in (\underline{\theta}_{CC}, \bar{\theta}_{CC}]$; offers with $x \leq \underline{\theta}_{CC}$ leave the contract unchanged. The value of CEO employment therefore satisfies:

$$\begin{aligned} \gamma V_C(r, \tau, f) = & \tilde{w}_C(r, \tau, f) + \frac{\partial V_C}{\partial \tau}(r, \tau, f) + \eta V_U \\ & + \lambda_1 \int_{\bar{\theta}_{CC}}^{\infty} \left[(1 - \beta)V_C(0, \tau, f) + \beta V_C(0, 0, x) - V_C(r, \tau, f) \right] d\Psi(x) \\ & + \lambda_1 \int_{\underline{\theta}_{CC}}^{\bar{\theta}_{CC}} \left[(1 - \beta)V_C(0, 0, x) + \beta V_C(0, \tau, f) - V_C(r, \tau, f) \right] d\Psi(x) \end{aligned} \quad (\text{A.20})$$

where retirement enters through the effective discount rate in γ and unattachment through ηV_U , with $V_U = V_E(0, 0, f_{min})$ replacing an unattached Bellman equation as in the main text.

A.5. Executive Value Function

The procedure for the executive value function is the same as before, though notation is slightly more cumbersome, as executives face three types of offers. By (A.16), the manager's gain upon an internal promotion opportunity is $\beta S(\tau, f)$ when the promotion is executed and zero otherwise, independently of her current contract, so internal promotion contributes the term $\lambda_0 \beta [S(\tau, f)]^+$, where $[x]^+ = \max\{x, 0\}$ enforces the acceptance condition (A.15). The value of executive employ-

ment satisfies:

$$\begin{aligned}
\gamma V_E(r, \tau, f) = & \tilde{w}_E(r, \tau, f) + \frac{\partial V_E}{\partial \tau}(r, \tau, f) + \eta V_U + \lambda_0 \beta [S(\tau, f)]^+ \\
& + \lambda_1 \int_{\bar{\theta}_{EC}}^{\infty} \left[(1 - \beta) V_E(0, \tau, f) + \beta V_C(0, 0, x) - V_E(r, \tau, f) \right] d\Psi(x) \\
& + \lambda_1 \int_{\underline{\theta}_{EC}}^{\bar{\theta}_{EC}} \left[(1 - \beta) V_C(0, 0, x) + \beta V_E(0, \tau, f) - V_E(r, \tau, f) \right] d\Psi(x) \quad (\text{A.21}) \\
& + \lambda_2 \int_{\bar{\theta}_{EE}}^{\infty} \left[(1 - \beta) V_E(0, \tau, f) + \beta V_E(0, 0, x) - V_E(r, \tau, f) \right] d\Psi(x) \\
& + \lambda_2 \int_{\underline{\theta}_{EE}}^{\bar{\theta}_{EE}} \left[(1 - \beta) V_E(0, 0, x) + \beta V_E(0, \tau, f) - V_E(r, \tau, f) \right] d\Psi(x)
\end{aligned}$$

A.6. Full-Extraction Values

Every bargaining object in Section A.2 is built from the values at $r = 0$. Setting $r = 0$ in (A.20) and (A.21) and noting that $\underline{\theta}_{ij}(0, \tau, f) = \bar{\theta}_{ij}(\tau, f)$ by Proposition 1, the renegotiation regions vanish, and the value functions evaluated at $r = 0$ are given by:

$$\begin{aligned}
\gamma V_C(0, \tau, f) = & \tilde{w}_C(0, \tau, f) + \frac{\partial V_C}{\partial \tau}(0, \tau, f) + \eta V_U + \lambda_1 \beta \int_{\bar{\theta}_{CC}}^{\infty} \left[V_C(0, 0, x) - V_C(0, \tau, f) \right] d\Psi(x) \quad (\text{A.22}) \\
\gamma V_E(0, \tau, f) = & \tilde{w}_E(0, \tau, f) + \frac{\partial V_E}{\partial \tau}(0, \tau, f) + \eta V_U + \lambda_0 \beta [S(\tau, f)]^+ \\
& + \lambda_1 \beta \int_{\bar{\theta}_{EC}}^{\infty} \left[V_C(0, 0, x) - V_E(0, \tau, f) \right] d\Psi(x) + \lambda_2 \beta \int_{\bar{\theta}_{EE}}^{\infty} \left[V_E(0, 0, x) - V_E(0, \tau, f) \right] d\Psi(x) \quad (\text{A.23})
\end{aligned}$$

Rearranging the option-value integrals via integration by parts and using (A.6) yields:

$$\int_{\bar{\theta}_{ij}}^{\infty} \left[V_j(0, 0, x) - V_i(0, \tau, f) \right] d\Psi(x) = \int_{\bar{\theta}_{ij}}^{\infty} \frac{\partial V_j}{\partial x}(0, 0, x) \bar{\Psi}(x) dx \quad (\text{A.24})$$

where the boundary term vanishes because the integrand is zero at the threshold by construction.

This is the representation used in computation.

A.7. The Renegotiation Window: Proofs

Throughout, write $\Phi_j(f') \equiv V_j(0, 0, f')$ for the poacher's full-extraction value, which is continuous and strictly increasing in f' by the Monotonicity step of the proof of Theorem 1. Recall from that proof that the thresholds satisfy:

$$\Phi_j(\bar{\theta}_{ij}) = V_i(0, \tau, f) \tag{A.25}$$

$$\Phi_j(\underline{\theta}_{ij}) = \frac{V_i(r, \tau, f) - \beta V_i(0, \tau, f)}{1 - \beta} \tag{A.26}$$

the second line rearranging (7). Write $D_i(r, \tau, f) \equiv V_i(0, \tau, f) - V_i(r, \tau, f)$, as in the main text. Where $V_i(0, \tau, f)$ or the right side of (A.26) falls outside the range of Φ_j , the corresponding threshold sits at an endpoint of the support of f' as in the proof of Theorem 1, and the monotonicity statements below hold with the same endpoint conventions.

Part (i) of the proposition follows from bounds on how tenure moves the value function at different contracts. The content of the lemma is that tenure gains are largest at full extraction and never fall below the bargaining share β of the full-extraction gain: at contracts below the ceiling, part of the match's growth accrues to the firm until competition forces it out.

Lemma 2 (Comparison bands). *Suppose $k(\cdot)$ is weakly increasing and, for the executive problem, that the promotion surplus $S(\tau, f)$ is weakly increasing in τ . Then for each type i , productivity f , contract $x \leq 0$, and tenures $\tau_1 \leq \tau_2$:*

- (a) $V_i(x, \tau, f)$ is strictly increasing in x ;
- (b) $G_i \equiv V_i(0, \tau_2, f) - V_i(0, \tau_1, f) \geq 0$; and
- (c) $\beta G_i \leq V_i(x, \tau_2, f) - V_i(x, \tau_1, f) \leq G_i$.

Proof. We prove the claim for the CEO problem and indicate at the end the modification for the executive problem. Let $\Lambda \equiv \lambda_1 + \eta + \mu$ be the total event rate. The value function is the unique

fixed point of the operator T defined by

$$(TV)(x, \tau, f) = \int_0^\infty \Lambda e^{-\Lambda s} \left[\int_0^s e^{-\delta u} (x + \kappa(\tau + u, f)) du + e^{-\delta s} Q_V(x, \tau + s, f) \right] ds,$$

where $\kappa(\tau, f) \equiv a + k(\tau) + f + \Delta_C$ collects the contract-invariant flow, and the event value is the mixture

$$Q_V(x, \tau, f) = \frac{\lambda_1}{\Lambda} \mathbb{E}_{f'} [P_V(x, \tau, f; f')] + \frac{\eta}{\Lambda} V_U + \frac{\mu}{\Lambda} \cdot 0,$$

with post-offer value defined directly from the bargaining outcomes of Theorem 1:

$$P_V(x, \tau, f; f') = \begin{cases} \max \{ V(x, \tau, f), (1 - \beta)\Phi(f') + \beta V(0, \tau, f) \}, & \Phi(f') \leq V(0, \tau, f), \\ (1 - \beta) V(0, \tau, f) + \beta \Phi(f'), & \Phi(f') > V(0, \tau, f), \end{cases}$$

where $\Phi(f') \equiv V(0, 0, f')$ is computed from the argument V itself. The max formulation makes P_V well defined for any candidate V ; when V has the threshold structure of Theorem 1, the two branches reproduce ignore/renege ($f' \leq \bar{\theta}$) and accept ($f' > \bar{\theta}$). On the relevant compact state space the flows are bounded, T maps bounded functions to bounded functions, and $\|TV - TV'\|_\infty \leq \frac{\Lambda}{\Lambda + \delta} \|V - V'\|_\infty$, so T is a contraction with the equilibrium value function as its unique fixed point.

Let $\mathcal{V}$ be the set of bounded functions that are weakly increasing in x , satisfy (b) and (c), and are weakly increasing in f . Strict monotonicity in x cannot be imposed on $\mathcal{V}$, because it is not preserved under sup-norm limits; it is recovered for the fixed point once $T\mathcal{V} \subseteq \mathcal{V}$ is established, since the flow term contributes $(x_2 - x_1) \int_0^\infty \Lambda e^{-\Lambda s} \int_0^s e^{-\delta u} du ds > 0$ while the event term is weakly increasing in x . $\mathcal{V}$ is nonempty and closed under sup-norm limits, so it suffices to show $T\mathcal{V} \subseteq \mathcal{V}$. Preservation of monotonicity in x and in f is routine (maxima and mixtures of monotone functions are monotone, and κ is increasing in f), as is (b): the flow increment $\int e^{-\delta u} [\kappa(\tau_2 + u, f) - \kappa(\tau_1 + u, f)]$ is nonnegative because k is weakly increasing, and each event branch of ΔP^0 is nonnegative (verified within the case analysis below). We verify (c).

Case analysis. Fix $V \in \mathcal{V}$, $x \leq 0$, $\tau_1 \leq \tau_2$, and write $V_m^0 \equiv V(0, \tau_m, f)$, $V_m^x \equiv V(x, \tau_m, f)$, $G \equiv V_2^0 - V_1^0 \geq 0$. Property (c) of the input gives $\beta G \leq V_2^x - V_1^x \leq G$. Define the input thresholds by $\Phi(\bar{\theta}_m) = V_m^0$ and $\Phi(\underline{\theta}_m) = \frac{V_m^x - \beta V_m^0}{1 - \beta}$; the input bands imply $\bar{\theta}_1 \leq \bar{\theta}_2$ (from $G \geq 0$) and $\underline{\theta}_1 \leq \underline{\theta}_2$ (from $V_2^x - \beta V_2^0 \geq V_1^x - \beta V_1^0$, the lower band). Write $W_m(f') \equiv (1 - \beta)\Phi(f') + \beta V_m^0$ and $A_m(f') \equiv (1 - \beta)V_m^0 + \beta\Phi(f')$ for the renegotiation and acceptance values, and note $W_2 - W_1 = \beta G$ and $A_2 - A_1 = (1 - \beta)G$. Let $\Delta P^x \equiv P(x, \tau_2; f') - P(x, \tau_1; f')$ and ΔP^0 likewise at $x = 0$. Because the flow increment is common to x and 0 and the exit values are constants, it suffices to show, pointwise in f' ,

$$\beta \Delta P^0 \leq \Delta P^x \leq \Delta P^0 \quad \text{and} \quad \Delta P^0 \geq 0.$$

There are five regions of f' .

1. $f' \leq \underline{\theta}_1$ (*ignored at both tenures*). $\Delta P^x = V_2^x - V_1^x \in [\beta G, G]$ and $\Delta P^0 = G$: both bounds hold by the input band.
2. $\underline{\theta}_1 < f' \leq \min\{\underline{\theta}_2, \bar{\theta}_1\}$ (*renegotiated at τ_1 , ignored at τ_2*). Here $\Delta P^0 = G$ and $\Delta P^x = V_2^x - W_1(f')$. Lower bound: $f' \leq \underline{\theta}_2$ implies $V_2^x \geq W_2(f')$, so $\Delta P^x \geq W_2 - W_1 = \beta G$. Upper bound: $f' > \underline{\theta}_1$ implies $W_1(f') \geq V_1^x$, so $\Delta P^x \leq V_2^x - V_1^x \leq G$.
3. $\underline{\theta}_2 < f' \leq \bar{\theta}_1$ (*renegotiated at both; arises when $\underline{\theta}_2 \leq \bar{\theta}_1$*). $\Delta P^x = W_2 - W_1 = \beta G$ and $\Delta P^0 = G$: both bounds hold.
4. $\bar{\theta}_1 < f' \leq \bar{\theta}_2$ (*accepted at τ_1 , retained at τ_2*). Here $P_1^x = P_1^0 = A_1(f')$, and $V_1^0 < \Phi(f') \leq V_2^0$, so

$$\Delta P^0 = V_2^0 - A_1(f') = G - \beta(\Phi(f') - V_1^0) \in [(1 - \beta)G, G] \geq 0.$$

- 4a. If $f' > \underline{\theta}_2$ (*renegotiated at τ_2*): $\Delta P^x = W_2(f') - A_1(f')$. Direct computation gives

$$\Delta P^0 - \Delta P^x = (1 - \beta)(V_2^0 - \Phi(f')) \geq 0, \quad \Delta P^x - \beta \Delta P^0 = (1 - \beta)^2(\Phi(f') - V_1^0) \geq 0,$$

using $\Phi(f') \leq V_2^0$ and $\Phi(f') > V_1^0$ respectively.

4b. If $f' \leq \underline{\theta}_2$ (ignored at τ_2 ; arises when $\bar{\theta}_1 < \underline{\theta}_2$): $P_2^x = V_2^x \geq W_2(f')$, so the lower bound follows from case 4a; and $P_2^x = V_2^x \leq V_2^0$ gives $\Delta P^x \leq V_2^0 - A_1 = \Delta P^0$.

5. $f' > \bar{\theta}_2$ (accepted at both tenures). $\Delta P^x = \Delta P^0 = A_2 - A_1 = (1 - \beta)G$: the upper bound binds with equality and the lower bound is $(1 - \beta)G \geq \beta(1 - \beta)G$.

In every case $\beta\Delta P^0 \leq \Delta P^x \leq \Delta P^0$ and $\Delta P^0 \geq 0$. Integrating over f' and s and adding the common flow increment $F \geq 0$: the upper bound gives $\Delta(TV)^x = F + \tilde{\lambda} \mathbb{E}[\Delta P^x] \leq F + \tilde{\lambda} \mathbb{E}[\Delta P^0] = \Delta(TV)^0$, and the lower bound gives $\Delta(TV)^x \geq \beta F + \beta \tilde{\lambda} \mathbb{E}[\Delta P^0] = \beta \Delta(TV)^0$ (using $F \geq \beta F$), where $\tilde{\lambda}$ collects the arrival weights. Hence TV satisfies (b) and (c), completing the induction for the CEO problem.

The executive problem. Executives face two cross-firm offer types (rates λ_1, λ_2), each of which admits the identical case analysis with $\Phi_{EC}(f') = V_C(0, 0, f')$ and $\Phi_{EE}(f') = V_E(0, 0, f')$, both tenure-free, in place of Φ , plus the internal-promotion event (rate λ_0). By (4) and the gate (5), the post-promotion value is $P^{\text{prom}}(x, \tau) = V_E(x, \tau, f) + \beta S^+(\tau, f)$ with $S^+ \equiv \max\{S, 0\}$, where we hold S fixed at its equilibrium value (the fixed point of the resulting operator is the equilibrium executive value function, since the two agree there). The promotion increment $\beta \Delta S^+$ is common to every contract x , including $x = 0$, so with $\Delta S^+ \geq 0$ (the hypothesis that S is weakly increasing in τ):

$$\Delta P^{\text{prom},x} - \beta \Delta P^{\text{prom},0} = (\Delta V_E^x - \beta G_E) + \beta(1 - \beta) \Delta S^+ \geq 0, \quad \Delta P^{\text{prom},x} \leq G_E + \beta \Delta S^+ = \Delta P^{\text{prom},0},$$

so the bands are preserved through the promotion channel as well. ■

Proof of Proposition 2. Part (i). For the acceptance threshold, the first equality of (A.26) gives $\Phi_j(\bar{\theta}_{ij}(\tau_2, f)) = V_i(0, \tau_2, f) \geq V_i(0, \tau_1, f) = \Phi_j(\bar{\theta}_{ij}(\tau_1, f))$ by Lemma 2(b), and strict monotonicity of Φ_j delivers $\bar{\theta}_{ij}(\tau_2, f) \geq \bar{\theta}_{ij}(\tau_1, f)$. For the renegotiation threshold, fix the contract $x = r$ between negotiations. The second equality of (A.26) gives

$$\Phi_j(\underline{\theta}_{ij}(r, \tau_2, f)) - \Phi_j(\underline{\theta}_{ij}(r, \tau_1, f)) = \frac{[V_i(r, \tau_2, f) - V_i(r, \tau_1, f)] - \beta G_i}{1 - \beta} \geq 0$$

by the lower band of Lemma 2(c), so $\underline{\theta}_{ij}$ is weakly increasing in τ .

Part (ii). Write $z = \Phi_j(f')$ for a consequential offer's deliverable value, $L \equiv \Phi_j(\underline{\theta}_{ij})$, and $H \equiv \Phi_j(\bar{\theta}_{ij})$. A renegotiation triggered by f' delivers the value (9); subtracting $V_i(r, \tau, f) = (1 - \beta)\Phi_j(\underline{\theta}_{ij}) + \beta V_i(0, \tau, f)$ (the second equality of (A.26)) gives the manager's value gain from the offer,

$$\text{Gain}(z) = \begin{cases} (1 - \beta)(z - L), & L < z \leq H \quad (\text{renegotiation}), \\ D_i + \beta(z - H), & z > H \quad (\text{move}), \end{cases} \quad (\text{A.27})$$

where the move branch follows from (8), since $V_i(0, \tau, f) - V_i(r, \tau, f) + \beta(\Phi_j(f') - V_i(0, \tau, f)) = D_i + \beta(z - H)$, and continuity at $z = H$ is immediate from the width identity, $D_i = (1 - \beta)(H - L)$. For fixed z , the renegotiation gain $(1 - \beta)(z - L)$ is weakly decreasing in τ because $L = \Phi_j(\underline{\theta}_{ij})$ is weakly increasing in τ by part (i). If the rising lower threshold passes z , the offer is ignored and its gain falls to zero.

Part (iii). An accepted offer's contract solves (8): $V_j(r', 0, f') = (1 - \beta) V_i(0, \tau, f) + \beta \Phi_j(f')$. The right side is weakly increasing in τ by Lemma 2(b), and $V_j(\cdot, 0, f')$ is strictly increasing in the contract by Lemma 2(a), so r' is weakly increasing in τ . ■

B. Additional Results

Figure B.1. Firm heterogeneity distribution

This figure displays the estimated position productivity distribution, which is characterized by location f_{min} , shape s_1 and scale s_2 from Table 3. In the figure, we display the (simulated) empirical CDFs of position-types, conditional on experience t .

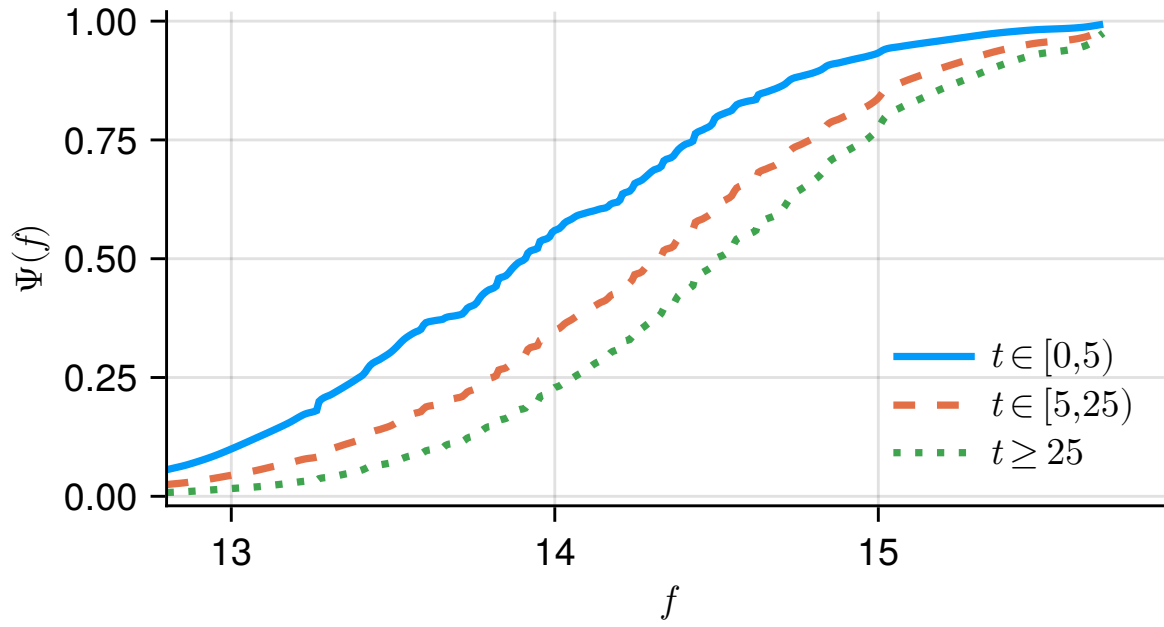


Table B.1. Managerial human capital composition

This table shows a decomposition of accumulated CEO human capital into general human capital $g(t)$, and firm-specific human capital $k(\tau)$ across labor market experience t and across tenure at a firm τ . We compute a variance decomposition: $s_x = \text{Cov}(h, x) / \text{Var}(h)$ for $x \in \{g, k\}$.

	(1)	(2)
	General human capital (g)	Firm-specific human capital (k)
Full Sample	51.04%	48.96%
By Role		
Executive	48.64%	51.36%
CEO	57.41%	42.59%
By Tenure		
$\tau \in [0, 5)$	57.27%	42.73%
$\tau \in [5, 10)$	82.64%	17.36%
$\tau \in [10, 25)$	93.76%	6.24%
By Experience		
$t \in [0, 5)$	12.86%	87.14%
$t \in [5, 10)$	11.24%	88.76%
$t \in [10, 25)$	54.55%	45.45%
By Firm Size		
Small	45.64%	54.36%
Medium	54.34%	45.66%
Large	55.00%	45.00%

Table B.2. Firm-Specific Human Capital and Executive Mobility

This table reports the labor market transitions of sitting executives in the baseline economy (firm-specific k) and in the counterfactual with no k -lock (portable k); the Diff columns report no-lock – baseline (in percentage points). Internal Promotion = promotion to CEO within the firm, External Promotion = promotion to CEO at another firm, and Horizontal Move = lateral executive move. Rows report the full sample and cuts by firm tenure, labor market experience, and firm size. It is the executive-side counterpart to Table 6, which reports how CEO appointments are filled, and is discussed in Section 7.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Internal Promotion			External Promotion			Horizontal Move		
	Base	No k -lock	Diff	Base	No k -lock	Diff	Base	No k -lock	Diff
Full Sample	63.83%	59.46%	-4.38 p.p.	8.15%	8.66%	+0.51 p.p.	28.01%	31.88%	+3.87 p.p.
By Tenure									
$\tau \in [0, 5)$	60.74%	59.17%	-1.57 p.p.	8.47%	8.69%	+0.22 p.p.	30.80%	32.15%	+1.35 p.p.
$\tau \in [5, 10)$	64.68%	58.83%	-5.85 p.p.	8.18%	9.07%	+0.89 p.p.	27.14%	32.10%	+4.95 p.p.
$\tau \in [10, 25)$	67.30%	60.25%	-7.05 p.p.	7.67%	8.21%	+0.55 p.p.	25.03%	31.54%	+6.51 p.p.
By Experience									
$t \in [0, 5)$	59.81%	58.31%	-1.50 p.p.	8.45%	8.71%	+0.26 p.p.	31.74%	32.98%	+1.24 p.p.
$t \in [5, 10)$	62.26%	57.60%	-4.67 p.p.	8.25%	8.82%	+0.57 p.p.	29.49%	33.58%	+4.09 p.p.
$t \in [10, 25)$	67.05%	60.85%	-6.20 p.p.	7.82%	8.54%	+0.71 p.p.	25.12%	30.61%	+5.48 p.p.
By Firm Size									
Small	47.22%	45.16%	-2.06 p.p.	9.36%	9.33%	-0.02 p.p.	43.42%	45.50%	+2.08 p.p.
Medium	60.66%	55.02%	-5.64 p.p.	9.85%	10.30%	+0.45 p.p.	29.49%	34.68%	+5.19 p.p.
Large	83.13%	77.37%	-5.77 p.p.	5.32%	6.39%	+1.07 p.p.	11.55%	16.24%	+4.69 p.p.

C. Estimation Appendix

C.1. Simulation Protocol

We simulate five independent panels of $I = 2,000$ managers for 100 years and discard the first 25 years as burn-in. Simulated moments are averaged across the five resulting 75-year panels.

At initialization, each manager draws ability $a \sim N(0, \sigma_a^2)$, is allocated between executive and CEO employment using the entry rates γ_E and γ_C , and begins with experience and tenure equal to zero. During each discretized period, separation, offer, and retirement events are drawn at the model's arrival rates. Offers draw a firm type from the relevant ability-tier distribution and are resolved using the equilibrium protocol of Section 3.2; unattached managers return through executive or CEO offers. A retiree is replaced by a new entrant, and retirement is imposed after 45 years of experience if it has not occurred earlier. The burn-in makes the post-burn-in moments insensitive to the initial allocation.

Firm types form a product grid of two mean-zero size levels and 50 quantiles of the size-orthogonal component ν , for 100 base types. The size grid has standard deviation $\sigma_s = 1.76$, matching the dispersion of log assets in the estimation sample. For ability tier q , offer probabilities over size levels are tilted according to (14); conditional on size, ν is drawn independently from (13). A manager retains the employer's type until leaving the firm, including through an internal promotion.

Tenure and experience are discretized on 51-point grids, with experience capped at 50 years, and the piece rate on a 50-point grid. Managers discount at 5% per year. We require the cubic profiles $g(t)$ and $k(\tau)$ to remain nonnegative on their computational grids. Because sorting affects the manager's problem only through the offer distribution, the value functions are solved once per ability tier.

The same random draws are used at every candidate parameter vector. Holding the sequence of ability, firm, and labor market shocks fixed across evaluations makes the simulated moments a deterministic function of the parameters, so that the objective in (C.5) is smooth in Θ and

differences between parameter vectors reflect the model rather than simulation noise. The same common random numbers are used for the finite-difference Jacobian described in Appendix C.6.

C.2. Construction of Hazard Moments

The retirement rate is calibrated separately. For the remaining mobility moments, we restrict the empirical panel to managers with at least one observed non-retirement transition. After simulating complete careers, we impose the same selection rule, spell definitions, and censoring conventions on the simulated panel before constructing its transition moments. The CEO employment-share target is calculated in this same selected sample.

We fit Kaplan–Meier estimates separately for each employment state and focal transition. For example, consider horizontal CEO transitions. Let $N_C(\tau)$ be the number of managers at risk at tenure τ and $M_C(\tau)$ the number of horizontal CEO transitions occurring at that tenure. The Kaplan–Meier estimate is

$$\hat{S}_C(\tau) = \prod_{s=0}^{\tau} \frac{N_C(s) - M_C(s)}{N_C(s)}.$$

Competing events are treated as right-censoring for each focal transition. Job-to-job moves are indexed by tenure, transitions into unattachment by experience, and re-employment by time in unattachment. We match $1 - \hat{S}(5)$ and $\hat{S}(5) - \hat{S}(10)$ for transitions between employment states and into unattachment, and $1 - \hat{S}(2)$ for re-employment. With competing events censored, these are selected-sample cause-specific KM auxiliary risks rather than population cumulative-incidence probabilities. The identical estimator is applied to the empirical and simulated panels.

C.3. Construction of the Fixed-Effect Moment Targets

Four of our targets are distributional moments of the AKM fixed effects estimated from (16): the mean, variance, and skewness of the firm effects $\zeta_{j(i,y)}$, and the variance of the manager effects ξ_i . Two features of how these are estimated require care before they can be matched to their model counterparts.

First, the empirical fixed effects are estimated with error. Managers move between firms

infrequently, so each firm effect is identified from a small number of movers and the estimated distribution of $\hat{\zeta}$ inherits substantial limited-mobility noise (Andrews et al., 2008; Kline et al., 2020). Writing $\hat{\zeta} = \zeta + e$ with e the estimation error, the plug-in variance and skewness of $\hat{\zeta}$ overstate those of ζ . This noise does not cancel against the simulated moment: in the simulation a firm is a productivity grid point with many managers attached, so the simulated fixed effects are recovered essentially without error. Matching the plug-in data moments to the simulated ones would therefore attribute estimation noise to genuine firm heterogeneity and inflate the estimated dispersion of f .

We instead target the signal content of these moments, estimated by split-sample covariance. Estimation errors are independent across disjoint random subsamples, so re-estimating (16) on two random halves A and B and taking the observation-weighted cross-product identifies the signal variance,

$$\hat{\text{Var}}(\zeta) = \overline{(\hat{\zeta}_A - \bar{\hat{\zeta}}_A)(\hat{\zeta}_B - \bar{\hat{\zeta}}_B)},$$

since the two error terms are uncorrelated with each other and with ζ . The analogous three-way split identifies the third central moment, and the same two-way splits deliver the manager-effect variance. We average over 20 random splits of each type. The correction is quantitatively important for the firm effects, removing a large share of the plug-in variance and reversing the sign of the plug-in skewness, and negligible for the manager-effect variance, which is estimated from many observations per manager. The mean firm effect and the two size slopes of Section 4.2.2 are unbiased under mean-zero estimation error and are not corrected.

Second, we winsorize the estimated fixed effects at the 0.5th and 99.5th percentiles before computing these moments, so that the targets are not driven by a small number of extreme estimated effects.

C.4. Weighting Matrix

From the empirical sample we obtain a $K \times 1$ vector of moments $\hat{M} \in \mathbb{R}^K$. Let $\psi_n \in \mathbb{R}^K$ denote the influence function of observation n on those moments. Observations are grouped by manager;

let $m \in \{1, \dots, \mathcal{M}\}$ indicate managers and $n \in m$ for the N_m observations which belong to manager m . We first stack the within-manager influence functions:

$$\Psi_m = \sum_{n \in m} \psi_n \in \mathbb{R}^K \quad (\text{C.1})$$

$$\Psi = [\Psi_1, \Psi_2, \dots, \Psi_{\mathcal{M}}]^T \quad (\text{C.2})$$

The clustered covariance matrix of the vector of moments can then be estimated as:

$$\text{avar}(\hat{M}) = \frac{1}{\mathcal{M}} \sum_{m=1}^{\mathcal{M}} \Psi_m \Psi_m^T \quad (\text{C.3})$$

The weighting matrix $\hat{W}$ is then obtained as the inverse of matrix (C.3). Let $\Theta \in \mathbb{R}^P$ denote an arbitrary vector of structural parameters. Define the moment residual $g : \mathbb{R}^P \rightarrow \mathbb{R}^K$ as:

$$g(\Theta) = \hat{M} - \frac{1}{S} \sum_{s=1}^S \hat{m}^s(\Theta) \quad (\text{C.4})$$

Where $\hat{M}$ is the vector of empirical moments, $\hat{m}^s(\Theta)$ is the vector of simulated moments given parameter values Θ in simulation s , and S is the total number of simulations. In our application $K = 33$, $P = 22$, and $S = 5$. The vector of estimates $\hat{\Theta}$ minimizes the SMM objective function:

$$\hat{\Theta} = \arg \min_{\Theta} g(\Theta) \hat{W} g(\Theta)' \quad (\text{C.5})$$

C.5. Model Estimation Algorithm

We minimize the SMM quadratic form using the TikTak algorithm. The routine proceeds as follows:

1. *Set initial guesses for model parameters:* We set bounds for each parameter and draw a sequence of 100,000 Sobol points from the resulting bounded parameter space. We then simulate the model, construct our simulated panel, and compute the SMM objective at each

Sobol point.

2. *Parallel local minimization:* We then rank the Sobol points by objective function value, and compute the local minimum around the top 225 points (225 is motivated by core usage on the cluster we use). We compute local minima using the [Subplex algorithm via NLOpt](#) (Rowan, 1990). We use 500 iterations for each of the 225 local minima, as we do not need to find the exact local minimum, rather just be in the neighborhood. We rank these 225 local minima by objective function value.
3. *Parallel global minimization:* We then compute convex combinations of each local minima with the current best point: for $i \dots N_S, i > 1$, we set $\theta_2 = 0.995$ and $\theta_{N_S} = 0.005$ as the maximum and minimum weight on the current best local minimum and the remaining Sobol points (ranked 2 to 225). We then compute the local minimum of each candidate global minimum point, using 1000 iterations of the Subplex algorithm. If $i^* > 1$, we update to the position of the best possible global minimum point and repeat the parallel global minimization by re-computing the convex combination of the current best candidate and all remaining points. We do this until i^* is stable between iterations

C.6. Standard Errors for Parameter Estimates

For true parameter vector Θ and consistent estimate $\hat{\Theta}$, we have the following asymptotic distribution (Duffie and Singleton, 1993):

$$\sqrt{\mathcal{M}}(\hat{\Theta} - \Theta) \rightarrow^d N(0, \text{avar}(\hat{\Theta})) \quad (\text{C.6})$$

where $\mathcal{M}$ is the number of managers, the level at which the moments are clustered. $\text{avar}(\hat{\Theta})$ can be expressed as:

$$\text{avar}(\hat{\Theta}) = \left(1 + \frac{1}{S}\right) \left(\frac{\partial g(\Theta)}{\partial \Theta} W \frac{\partial g(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{C.7})$$

where $\frac{\partial g(\Theta)}{\partial \Theta}$ is the Jacobian of the moment residual (C.4) with respect to the structural parameters, W is the optimal weighting matrix, and S is the number of simulations. We approximate the

Jacobian by central differences,

$$\frac{\partial \hat{g}_m(\Theta)}{\partial \Theta_p} = \frac{g_m(\hat{\Theta} + h_p e_p) - g_m(\hat{\Theta} - h_p e_p)}{2h_p}, \quad (\text{C.8})$$

for each moment m and parameter p , where e_p is the p th unit vector and the perturbation size is $h_p = \max\{0.05 |\hat{\Theta}_p|, \underline{h}\}$, five percent of the absolute value of the parameter estimate subject to a small floor.

Because each evaluation of g rests on a finite simulated sample, a Jacobian computed from a single set of draws inherits simulation noise, and that noise does not vanish as the perturbation shrinks. We therefore compute (C.8) independently under 30 different seeds, each drawing its own five panels of $I = 2,000$ managers, and average the resulting Jacobians. Within each seed the perturbed and baseline evaluations share the same draws, as described in Appendix C.1, so the differencing removes the common simulation error and the averaging across seeds removes what remains. The resulting standard errors stabilize well before the last seed is added: those computed from the first 20 seeds differ from the reported 30-seed values by less than three percent for every parameter, with a median difference of under one percent. The standard errors are the square root of the diagonal elements of the matrix:

$$\left(1 + \frac{1}{S}\right) \left(\frac{\partial \hat{g}(\Theta)}{\partial \Theta} \hat{W} \frac{\partial \hat{g}(\Theta)}{\partial \Theta'} \right)^{-1} \quad (\text{C.9})$$

where $\hat{W}$ is the sample counterpart of the optimal weighting matrix. One parameter, the CEO offer arrival rate γ_C , is estimated at the lower edge of its admissible range. Its reported standard error should be read with the usual caution that applies to inference at a boundary; none of our results turn on the precision of this parameter.

C.7. Statistical Identification

C.7.1. Parameter–Moment Sensitivity

To verify that the mapping described in Section 4.2 holds statistically, we compute the Andrews–Gentzkow–Shapiro sensitivity of the estimator to the target moments (Andrews et al., 2017), $\Lambda = -(G'WG)^{-1}G'W$, at the estimated parameter vector, standardized so that each entry measures the response of a parameter, in units of its asymptotic standard deviation, to a one-standard-deviation perturbation of a moment. Table C.1 reports the matrix; reading down a parameter’s column shows which moments identify it. The discussion below shows that this identification is also economically interpretable, following the three sets of moments of Section 4.2.

The labor market mobility moments identify the mobility and unattachment parameters essentially one hazard per parameter: η is read by the unattachment hazards (−0.68), λ_0 by the promotion hazards (−0.61), λ_1 by the external CEO hazards (−0.60), λ_2 by the executive-move hazards (−0.74), γ_E by the unattachment-to-executive entry hazard (−0.98), and γ_C by the CEO employment share (−0.87). Apart from the executive-move hazard at the five-to-ten year horizon, which the pay-distribution parameters also shift, no parameter outside the block loads comparably on these moments.

Within the Mincer wage regression, the tenure and experience profiles are separately identified, as the reset of tenure at job switches breaks the coupling between experience and tenure in the data. As Table C.1 shows, the level experience coefficients are read only by the reduced-form general profile, and the firm-specific coefficients load below 0.10 on each of them. Moreover, in first differences experience growth is absorbed into the constant, which leaves the first-difference tenure coefficients free of general-profile content.

The firm-fixed-effect moments discipline the pay-distribution block— f_{min} , s_1 , s_2 , Δ_C , and b_{size} —whose members all shift the firm-effect mean and skewness, CEO wage growth, and late-horizon mobility, and are therefore the harder case. Three features separate them. First, dedicated moments break into the block: the CEO indicator is read mainly by Δ_C , and the firm-effect size slope mainly by b_{size} ; likewise the manager-effect size slope is read almost exclusively by ρ_{sort} (−0.90),

since only sorting of ability onto firm size can move it. Second, location and dispersion load with opposite signs: f_{min} enters every shared moment with the opposite sign to s_1 and s_2 , so the data distinguish where the productivity distribution sits from how spread out it is. Among the remaining parameters, σ_a is read by the manager-effect variance (-0.67), while σ_e and ν_e are separated within the wage-residual block, using both the level and wage-growth residuals, by the usual variance-versus-autocovariance logic for an autoregressive process.

C.7.2. Statistical Identification of the Bargaining Power Parameter

Our identification argument for bargaining power comes from the level-versus-growth comparison of Section 4.2.2. Intuitively the concern it must overcome is the following: even conditional on controlling firm-wage premia and experience effects, wage levels may rise with tenure either through firm-specific human capital accumulation or renegotiated piece-rate gains, while wage growth responds to the piece rate only through its current revisions (i.e., human capital returns enter similarly in both the level and growth auxiliary wage equations). To make this precise, let $R^L(T)$ and $R^\Delta(T)$ denote the cumulative returns to tenure at horizon T implied by the tenure profiles of the level and growth regressions. In the model, both decompose as

$$R^L(T) = K(T) + \Pi^L(T), \quad R^\Delta(T) = K(T) + \Pi^\Delta(T),$$

where $K(T)$ is the structural firm-specific human capital return, common to the two equations, and Π^L , Π^Δ are the equilibrium projections of the omitted piece rate onto tenure (i.e., the omitted variable biases introduced by not observing the piece-rate). By construction, the difference between these two measures of returns is independent of human capital returns:

$$\mathcal{W}(T) \equiv R^L(T) - R^\Delta(T) = \Pi^L(T) - \Pi^\Delta(T),$$

and our identification argument hinges on $\mathcal{W}(T)$ responding strongly to β , but weakly to other parameters, particularly human capital accumulation.

We use the sensitivity matrix and the delta method to develop asymptotic values for $\partial\mathcal{W}(T)/\partial\beta$ and for its counterpart with respect to firm-specific human capital. Because $K(T)$ depends on the three coefficients of $k(\tau)$ jointly, we perturb them proportionally: κ multiplies the estimated profile, $k(\tau; \kappa) = \kappa \hat{k}(\tau)$, so that $\kappa = 1$ at the estimate and a unit change in κ moves the entire firm-specific return by its estimated amount; κ_g is defined analogously on the general profile $\hat{g}(t)$. At a ten-year horizon, a one-standard-error movement in β shifts the wedge by -0.033 ; the corresponding shifts from the human capital scales are -0.001 (κ) and -0.003 (κ_g), even though κ moves the level and growth returns themselves as strongly as β does. Figure C.1 displays these derivatives.

The sensitivity matrix confirms that this identifying variation is β 's own, and not borrowed from the firm-heterogeneity and CEO-premium parameters whose pay-distribution moments $\hat{\beta}$ shares. In Table C.1, $\hat{\beta}$'s loadings on the level and first-difference quadratic tenure coefficients are nearly equal and opposite ($+0.15$ and -0.19)—the wedge pattern—with no loading on any level experience coefficient. To isolate what moves $\hat{\beta}$ but not the block, we orthogonalize $\hat{\beta}$'s row of the sensitivity matrix against the rows of f_{min} , s_1 , s_2 , Δ_C , and b_{size} : the resulting moment direction—one that moves $\hat{\beta}$ while leaving all five estimates unchanged—loads on firm-effect skewness against the firm-effect mean ($+0.64$ against -0.56), together with the CEO share ($+0.26$) and the CEO indicator (-0.27). The economics is transparent: a change in the location or dispersion of firm productivity moves the mean and the skewness of pay together, and the CEO premium is pinned by its dedicated indicator, while bargaining rents accrue unevenly across positions and twist the skewness of pay relative to its mean. β therefore retains its own identifying variation conditional on the entire pay-distribution block.

Figure C.1. Identification of Managerial Bargaining Power

This figure displays the response of the tenure returns implied by the level and wage growth auxiliary regressions to the structural parameters governing managerial bargaining power and firm-specific human capital accumulation. $R^L(T) = K(T) + \Pi^L(T)$ is the fitted cumulative tenure profile of the level regression (16) evaluated at $T = 10$ years of tenure, constructed from the three tenure coefficients only: experience terms, CEO indicators, and firm and manager effects are partialled out as controls. $R^\Delta(T) = K(T) + \Pi^\Delta(T)$ applies the same construction to the Δ tenure coefficients of the growth regression (17), cumulating the fitted within-job growth profile to a T -year return in the same units, with

$$\mathcal{W}(T) \equiv R^L(T) - R^\Delta(T) = \Pi^L(T) - \Pi^\Delta(T)$$

the difference in fitted level and growth returns up to tenure T . Here $K(T)$ is the structural firm-specific human capital return, which enters the two regressions identically, while $\Pi^L(T)$ and $\Pi^\Delta(T)$ are the components of the fitted returns attributable to the omitted negotiated piece rate: $\Pi^L(T)$ captures the accumulated stock of bargaining rents that loads onto tenure in wage levels, and $\Pi^\Delta(T)$ the contemporaneous piece-rate revisions that load onto tenure in wage growth. The common $K(T)$ cancels in $\mathcal{W}(T)$, so the difference isolates the bargaining components. We compute $\frac{\partial R^L(T)}{\partial \beta}$, $\frac{\partial R^\Delta(T)}{\partial \beta}$, and $\frac{\partial \mathcal{W}(T)}{\partial \beta}$ by central finite differences around the estimated parameter vector: β is perturbed in proportion to its estimate, and at each perturbed value the model is re-solved, a panel is re-simulated under common random numbers (identical shock draws across parameter values), and both regressions are re-estimated on the simulated panel. The same experiment for firm-specific human capital requires one additional step, because $k(\tau)$ is a profile governed by three coefficients rather than a single parameter. We introduce a scaling factor κ that multiplies the entire estimated profile, $k(\tau; \kappa) = \kappa \hat{k}(\tau)$, so that $\kappa = 1$ reproduces the estimate. Perturbing κ around one scales the whole firm-specific human capital profile up and down while preserving its shape, and we perturb κ exactly as we perturb β . To place the two derivatives on a common scale, bars report them in units of one standard error of each parameter. $\text{s.e.}(\beta) = 0.0415$ is the SMM standard error reported in Table 3. $\text{s.e.}(\kappa)$ is induced from the estimated profile coefficients by the delta method: writing $\hat{c} = (\hat{c}_1^k, \hat{c}_2^k, \hat{c}_3^k)'$ and Σ_c for their block of the SMM asymptotic covariance matrix, the scale of a coefficient vector c relative to the estimated profile is its projection $\kappa(c) = \hat{c}'c/\hat{c}'\hat{c}$, with $\kappa(\hat{c}) = 1$; since $\kappa(c)$ is linear in c , $\text{s.e.}(\kappa) = \sqrt{\hat{c}'\Sigma_c\hat{c}}/\hat{c}'\hat{c} = 0.049$.

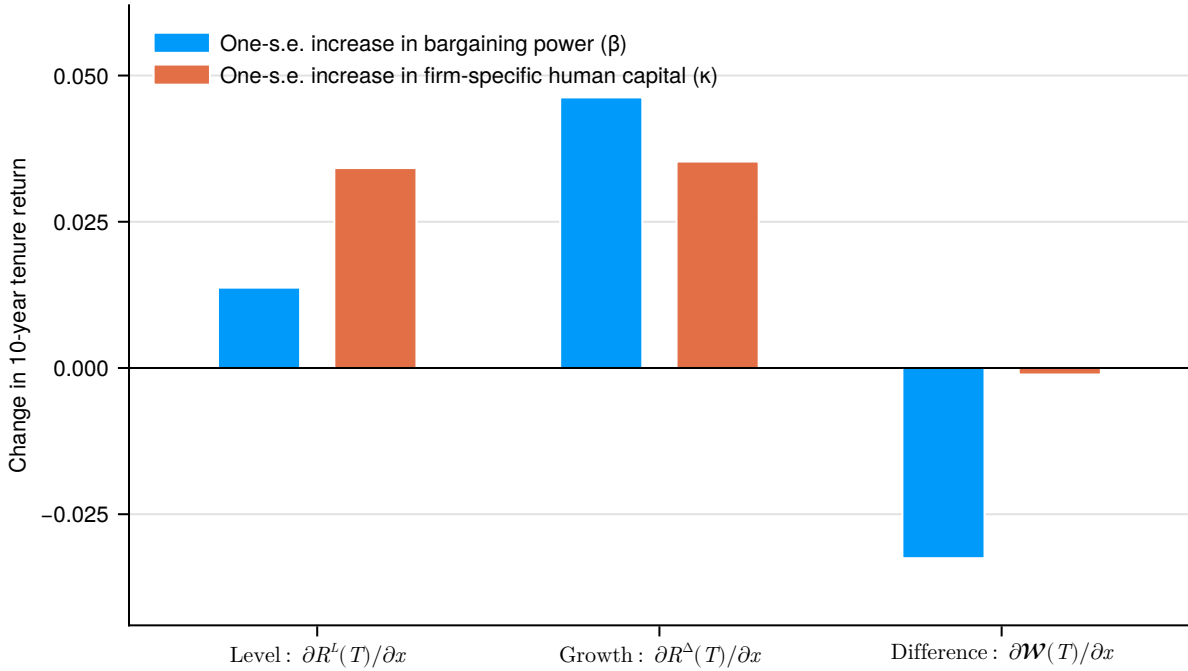


Table C.1. Sensitivity of Parameter Estimates to Target Moments

This table reports the standardized sensitivity of the estimated parameter vector to the target moments, following [Andrews et al. \(2017\)](#). Each entry gives the response of the column parameter, in units of its standard error, to a one-standard-deviation perturbation of the row moment; entries below 0.10 in absolute value are blanked for readability. Reading down a parameter's column shows which moments carry its identifying information. Rows and columns are ordered so that each block of moments is aligned with the block of parameters it disciplines. Panel A reports the labor market transition moments; Panel B reports the remaining moments. Moment numbering is continuous across panels.

[illegible]

Panel B: Wage, firm, and reduced-form moments

Moment	λ_0	λ_1	λ_2	η	γ_E	γ_C	β	p_{min}	s_1	s_2	δ_C	b_{size}	ρ_{sort}	σ_a	σ_e	v_e	c_1^k	c_2^k	c_3^k	c_1^g	c_2^g	c_3^g
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)
Firm fixed effect moments																						
(12) Firm effect (mean)		0.18	0.18				0.34	-0.40	0.37	0.32	-0.36	-0.34	-0.12		-0.19	-0.14	-0.19			-0.12	0.18	-0.17
(13) Firm effect (variance)		-0.43	0.15				-0.27	0.21	-0.26	0.10	0.21	-0.15	0.23	-0.19								
(14) Firm effect (skewness)		0.22	0.17				0.39	-0.35	0.41	0.36	-0.41	-0.29			-0.18	-0.17	-0.19				0.15	-0.13
CEO moments																						
(15) CEO indicator		0.23					0.15	-0.17	0.18	0.20	-0.39	-0.20					-0.13				0.11	-0.10
(16) CEO growth			-0.22				-0.50	0.52	-0.45	-0.49	0.35	0.48			0.12		0.26				-0.18	0.19
Firm size and sorting moments																						
(17) Firm effect (size slope)									0.11			-0.29		0.11								
(18) Manager effect (size slope)													-0.90	-0.35								
Wage dispersion moments																						
(19) Manager effect (variance)													0.12	-0.67						0.11	-0.10	
(20) Wage residual (variance)			-0.10				0.14		-0.22	-0.18	0.15	0.14	-0.15	0.45	0.37	0.46	-0.28	0.36	-0.30	0.49	-0.37	0.28
(21) Wage residual (autocovariance)							-0.11		0.15	0.12	-0.11		0.17	-0.22	-0.23	-0.26	0.22	-0.28	0.24	-0.37	0.29	-0.23
(22) Wage growth residual (variance)			-0.11				-0.13	0.17	-0.22	-0.23	0.22	0.21			0.40	0.61	0.11					
(23) Wage growth residual (autocovariance)							-0.15							-0.18	-0.63	-0.36				-0.25	0.18	-0.14
Tenure/Experience reduced-form coefficients																						
(24) Tenure (linear)							0.23	-0.22	0.15	0.17	-0.18	-0.17		0.10		0.11	-0.73	0.66	-0.61	0.37	-0.19	0.19
(25) Tenure (quadratic)							0.15	-0.15	0.12	0.14	-0.16	-0.13						-0.47	0.53		0.30	-0.32
(26) Tenure (cubic)																			-0.18			
(27) Tenure growth (quadratic)							-0.19	0.18	-0.11	-0.12	0.12	0.14		-0.12		-0.12	0.20	-0.23	0.24			
(28) Tenure growth (cubic)														0.15	0.18	0.20		0.11	-0.15	0.12		
(29) Experience (linear)																				-0.54	0.54	-0.54
(30) Experience (quadratic)																					-0.34	0.46
(31) Experience (cubic)																						-0.13
(32) Experience growth (quadratic)				-0.12			0.21	-0.19		0.12		-0.12		0.14	0.12	0.16	0.12	-0.20	0.19			
(33) Experience growth (cubic)			-0.12																-0.11			

C.8. The Welfare Effect of Job Lock on Managers

Log compensation decomposes as $w_{it} = \ell_{it} + v_{it}$, where $\ell_{it} = a_i + g(t) + k(\tau_{it})$ and

$$v_{it} = r_{it} + f_{it} + \Delta_C \mathbf{1}\{\text{CEO}\}_{it}, \quad (\text{C.10})$$

where r is the log piece rate, f is employer productivity, and Δ_C is the CEO premium. To isolate non-portability from the productive value of firm-specific capital, we compare the baseline with a counterfactual in which k is portable across firms but its productive contribution is unchanged. We exclude ℓ and measure the effect through v : contracts (r), employer allocation (f), and CEO status (Δ_C).¹

C.8.1. The Welfare Object

Because w is log compensation, flow utility equals w . Let $s = (r, \tau, f, i)$ denote the payoff-relevant state and π the equilibrium policy. The discounted value of each component x of v satisfies

$$\delta V^x(s) = x(s) + \mathcal{L}^\pi V^x(s), \quad x \in \{r, f, \Delta_C \mathbf{1}\{\text{CEO}\}\}, \quad (\text{C.11})$$

where $\mathcal{L}^\pi$ is the generator for tenure drift and promotion, offer, separation, and retirement transitions. Linearity implies $V^v = V^r + V^f + V^{\Delta_C}$. Let $H(s)$ denote expected discounted attached duration, satisfying $\delta H = \mathbf{1}\{s \text{ attached}\} + \mathcal{L}^\pi H$. Because attachment transitions are exogenous and common across regimes, so is H .

Define the welfare effect of job lock as $\Delta V^v(s) = V_{\text{base}}^v(s) - V_{\text{lock-off}}^v(s)$. Following [Lucas \(1987\)](#), the consumption-equivalent $\text{CE}(s)$ scales consumption in the lock-off regime to equate lifetime

¹For computation, we use an equivalent “destroy- k ” counterfactual that removes k from output. The portable- k and destroy- k regimes have identical decision rules because k is absent from comparisons across alternatives. Under common random numbers they therefore generate identical v -paths, so $\Delta v_{\text{base-destroy}} = \Delta v_{\text{base-portable}}$ channel by channel. The excluded component ℓ contains the direct productive value of k .

utility across regimes. Under log utility, $\log(1 + \text{CE})H(s) = \Delta V^v(s)$, so

$$\text{CE}(s) = \exp\left(\frac{\Delta V^v(s)}{H(s)}\right) - 1, \quad \text{CE}^x(s) = \text{CE}(s) \frac{\Delta V^x(s)}{\Delta V^v(s)}. \quad (\text{C.12})$$

The second expression allocates the total to surplus capture, employer productivity, and the CEO premium in proportion to their contributions to ΔV^v ; the components sum to $\text{CE}(s)$.² A positive (negative) value means job lock raises (lowers) manager welfare. Table 8 reports the total and its components.

C.8.2. Estimation

We estimate the channel values by simulating careers under the equilibrium policy π . The baseline and lock-off simulations use common random numbers for ability, arrivals, and productivity. At each baseline node t , the realized discounted values are

$$\hat{V}_i^x(t) = \sum_{q \geq t} e^{-\delta(q-t)} x_i(q), \quad \hat{H}_i(t) = \sum_{q \geq t} e^{-\delta(q-t)}, \quad (\text{C.13})$$

where q indexes subsequent attached annual observations. These finite-horizon sums approximate the continuous-time values up to a common discretization factor that cancels below. For matched simulations, let $\Delta \hat{V}_i^x(t)$ denote the baseline-minus-lock-off difference. For baseline states in group $\mathcal{G}$, the reported statistic and decomposition are

$$\text{CE}(\mathcal{G}) = \exp\left(\frac{\sum_{(i,t) \in \mathcal{G}} \Delta \hat{V}_i^v(t)}{\sum_{(i,t) \in \mathcal{G}} \hat{H}_i(t)}\right) - 1, \quad \text{CE}^x(\mathcal{G}) = \text{CE}(\mathcal{G}) \frac{\sum_{\mathcal{G}} \Delta \hat{V}^x}{\sum_{\mathcal{G}} \Delta \hat{V}^v}. \quad (\text{C.14})$$

We aggregate value differences and discounted horizons before applying the consumption-equivalent transformation. The sample consists of attached, market-exposed baseline states. Continuation windows may overlap, so inference clusters by manager.

²Because $x \mapsto e^x - 1$ is nonlinear, these are additive components of the total consumption-equivalent, not standalone channel consumption-equivalents.